Software Requirements Specification

Dawra Campus

Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Version 1.0 – Phase 3 Requirements Document

Prepared by Group 4: Sultan Almheiri (100065654), Zayed Alfadli (100064657), Ghaith Alhinaai (100066185)

Khalifa University, Department of Computer Science

COSC 336 Introduction to Software Engineering, Lab Section B2

Prepared for the lab instructors Ala' Alsamarneh and Hamdan Abdulla Naser Salem Alshkeili

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Revision History

Name	Date	Reason for changes	Version
Group 4	15 Sep 2026	First requirements, in the Phase 1 planning document	0.1
Group 4	29 Sep 2026	Scope, roles and tools confirmed in the Phase 2 feasibility study	0.2
Group 4	13 Oct 2026	Full SRS for Phase 3: use cases, states, business rules and detailed requirements	1.0

1. Introduction

1.1 Purpose

This is the Software Requirements Specification (SRS) for Dawra Campus, the campus asset-reuse system our group is building for COSC 336. It lists what the system has to do, so we can design it in Phases 4 and 5, build it in Phase 6 and test it in Phase 7.

It covers the whole prototype: the normal (classic) features such as accounts, assets, requests, approvals, transfers and maintenance; the AI features; and the sustainability numbers. It builds on the requirements we wrote in Phase 1 and the decisions we made in the Phase 2 feasibility study. It says what the system should do, not how we will build it.

1.2 Document Conventions

We follow the SRS template from the project description. Every requirement has an ID. If a requirement comes from Phase 1, it keeps the Phase 1 number, so FR-07.3 is the third detailed requirement under Phase 1's FR-07 (resource requests).

ID	What it is
FR-nn.m	Classic functional requirement (sections 4.3–4.16)
AI-nn.m	AI requirement (sections 4.17–4.23)
SU-nn	Sustainability requirement (section 6.1)
NFR-nn	Non-functional requirement (section 5)
BR-nn	Business rule (section 5.5)
UC-nn, UI-nn, SI-nn	Use case (4.1), screen (3.1), software interface (3.3)
AS, LS, RQ, RV, TR, MW, EL, AT	States of assets, listings, requests, reservations, transfers, maintenance jobs, end-of-life actions and approvals (section 4.2)
TBD-n	Something we still need the instructors to confirm (Appendix C)

How requirements are written. "Shall" means the system must do it. We use the EARS patterns from the Chapter 3 slides: *The system shall …*, *When …*, *While …*, *If … then …* and *Where …*.

Priorities. The project description asks us to split requirements into essential prototype functions, desirable extensions and future enhancements. The table shows how these match the Must/Should/Could labels we used in Phase 1 and the High/Medium/Low labels in the template. Each feature has a priority, and its requirements share it unless a row says otherwise.

Course tier	Phase 1 label	Template	Meaning
Essential prototype function	Must	High	Has to work in the Phase 6 prototype and the final demo
Desirable extension	Should	Medium	We build it once the essential parts work
Future enhancement	Could / Won't	Low	Not this semester unless we finish everything else

Numbers and data. The numbers here are targets, not results, because nothing is built yet. All our test data is made up (synthetic). Anything that depends on KU policy is marked TBD and listed in Appendix C.

1.3 Intended Audience and Reading Suggestions

Reader	Why they read it	Where to start
Lab instructors (our client)	Check the scope, priorities and open questions	1.4, 2, 4.1, 6.4, Appendix C
Our team as developers	Design and build the system	2, 4.2, 5.5, then 4.3–4.23 and 5.1–5.4
Our team as testers (Phase 7)	Write test cases; every "shall" is something to test	4.2, 4.3–4.23, 5, 6.1
Whoever designs the screens	Plan the pages and user journeys	2.3, 3.1, 4.1

Section 2 describes the product. Section 3 covers its interfaces. Section 4 has the use cases, the states and the requirements for each feature. Section 5 has the quality requirements and the business rules, including who approves what. Section 6 covers sustainability, data, priorities and traceability. The appendices have the glossary, the open questions and our AI-use note.

1.4 Product Scope

Departments at KU keep buying things like chairs, monitors and lab equipment while the same items sit unused in another building. The asset register records who owns what, but it doesn't help anyone find, match or move these items (Phase 1, section 1.2).

Dawra Campus is a website for KU staff that fixes this. Departments register their assets and list the ones they don't need. Other departments search, request and reserve items, and the right people approve the transfer. The system also tracks repairs and what happens to items at the end of their life, and reports how much money and waste was saved. AI helps sort items into categories, match requests to items, suggest what to do with old items, answer questions and summarise reports, but people always make the final decision.

Our goals are the same as in Phase 1: fewer unnecessary purchases, longer asset life, better matching, clear approvals, measurable sustainability results, responsible AI, and a comparison of classic and AI-based approaches. What's left out is the same as in Phase 2: connecting to KU's real finance, procurement or ERP systems, payments, barcode or RFID scanners, training our own AI models, AI approving things on its own, and phone apps. This is a student prototype. Using it for real at KU would need proper evidence and a decision from the university.

1.5 References

ID	Document	Date	Location
R1	COSC 336 Lab-based Running Project – Project Description, including the SRS template in the appendix	Fall 2026	docs/reference
R2	COSC 336 Lab Phase 1 slides – Initial Plan and Requirement Gathering	Fall 2026	docs/reference
R3	Group 4, Phase 1 Initial Plan and Requirement Gathering Document	15 Sep 2026	docs/phase1
R4	Group 4, Dawra Campus Phase 2 Feasibility Study	29 Sep 2026	docs/phase2/final
R5	COSC 336 lecture slides, Chapter 3 Requirements Engineering (use cases, UML use-case diagrams, EARS)	Fall 2026	Blackboard
R6	ISO/IEC/IEEE 29148:2018 Systems and software engineering – Requirements engineering	2018	iso.org
R7	A. Mavin et al., "Easy Approach to Requirements Syntax (EARS)", IEEE International Requirements Engineering Conference	2009	IEEE Xplore
R8	I. Sommerville, Software Engineering, 10th edition, Pearson, chapter 4	2016	Course reference

2. Overall Description

2.1 Product Perspective

Dawra Campus is a new system. It doesn't replace KU's current asset register; it runs next to it. When an item is transferred or disposed of in Dawra Campus, staff can export the record and update the official register themselves. There is no direct connection to KU's systems in this version (Phase 2, section 5).

The table shows who and what sits outside the system and how they connect to it. We will draw the full context diagram in Phase 4.

Outside the system	How it connects
Staff in the nine roles	Use the website in a browser (section 2.3)
AI service provider	Gets only the fields it needs for a task and sends back suggestions, scores, explanations and draft text through one AI adapter (sections 3.3 and 4.17–4.23)
KU's current asset register	Not connected. Staff update it using the transfer and end-of-life records we export (FR-11.7)
Finance, Procurement and EHS offices	Take part through the Finance officer, Procurement officer and Sustainability officer roles, and set the policy values in Appendix C
Hosting and database provider	Runs the website, database and file storage (section 2.4)

Inside, the system is split into these parts, covered one by one in sections 4.3 to 4.23: accounts and access; departments and settings; asset records; listings, requests, search and reservations; approvals; transfers; inspections and maintenance; end-of-life actions; history, audit and notifications; reports and sustainability; and the AI services.

2.2 Product Functions

Classic features

- Create user accounts and give each person roles for their department.
- Register assets with their details, photos and documents.
- List unused items once the owning department approves.
- Submit, track, change and cancel requests, and search the listed items.
- Reserve items for a limited time without the same item going to two people.
- Send every approval to the right person, with a reason for any rejection.
- Record transfers from approval to handover and receipt, updating the owner and location.
- Record inspections, repairs and their costs.
- Propose, approve and record donation, recycling, retirement and disposal.
- Show each asset's full history, keep a log that can't be edited, send notifications and produce reports.

AI features

- Suggest a category, tags and a description when an asset is registered.
- Match requests to suitable items by meaning, after the normal rules remove items that don't qualify, with a score and an explanation.
- Suggest what to do with an old item and explain why.
- Answer questions and help draft requests through a chat assistant, always asking before changing anything.
- Write report summaries using the numbers from the normal reports.
- Record which AI model produced each result and whether the user accepted it, and fall back to the normal features if AI isn't available.

Sustainability

- Calculate waste diverted, avoided purchases, reuse rate, repair success, extra asset life, money saved and estimated carbon saved.

2.3 User Classes and Characteristics

Dawra Campus has nine staff roles. One person may hold several roles, with each assignment limited to named departments. The active role and department determine what that person can view and do; a second role does not remove the normal approval steps. Requesters, Asset custodians and Department representatives are the primary user classes because they carry out the everyday request and transfer process. The other classes provide policy review, specialist work or administration. The frequencies below are expected prototype use patterns, not measured KU usage.

User class	Main tasks in the system	Frequency of use	Technical expertise	Privilege level	Importance
Department representative	Review department needs; confirm asset records; approve listing release, requests, transfers and repair spending.	Daily or when a decision is needed	Basic browser and form skills; understands department responsibilities	Department records and assigned approval tasks	Primary
Asset custodian	Register and verify assets; prepare listings; confirm holds, handover and receipt; report defects and record completion evidence.	Daily	Basic browser skills and knowledge of asset condition and location	Operational records for assigned assets and departments	Primary
Requester	Search published listings; submit and track requests; review matches; request reservations and start transfers.	When a resource is needed	Basic browser and search skills	Own requests and linked records; authorised published listings	Primary
Business administrator	Maintain departments and categories; authorise role assignments; route exceptions; review operational reports and audit records.	Daily or weekly	Comfortable with administrative forms and business workflows	Business configuration and oversight within assigned departments	Secondary
Procurement officer	Review procurement-controlled transfers and relevant policy and reuse reports.	When policy approval or reporting is needed	Procurement-policy and record-review skills	Relevant records; transfer approval only when BR-06 applies	Secondary
Finance officer	Review threshold-controlled transfers and repair spending; approve end-of-life write-offs; review cost and savings reports.	When a financial decision is needed	Financial-review and reporting skills	Relevant financial records and assigned approval stages	Secondary
Maintenance staff	Inspect assigned assets; assess faults and repair costs; record approved work, parts, costs and resulting condition.	Daily or when work is assigned	Equipment inspection and maintenance skills; basic form skills	Assigned inspection and work-order records	Secondary
Sustainability officer	Review reuse-first end-of-life proposals, special-handling evidence and sustainability indicators.	When an action or report needs review	Sustainability and environmental-record skills	Relevant asset evidence; sustainability review and report review	Secondary
System administrator	Create and deactivate accounts; apply authorised role assignments; operate access, security and recovery settings.	During setup and as needed	Technical administration and security skills	Technical configuration; no automatic business approval rights	Secondary

2.3.1 Permission Matrix

Role	Assets	Listings	Requests	Reservations	Transfers	Maintenance	End-of-life	Users and roles
Department representative	C,V,E,A	C,V,E,A	V,A	V,E,A	V,A	C,V,A	C,V,E	-
Asset custodian	C,V,E,A	C,V,E	V	V,E,A	V,E	C,V,E	C,V,E	-
Requester	V	V	C,V,E	C,V,E	C,V	-	-	-
Business administrator	V	V	V	V	V	V	V	C,V,A
Procurement officer	V	V	V	V	V,A	-	V	-
Finance officer	V	V	V	V	V,A	V,A	V,A	-
Maintenance staff	V	V	-	-	-	C,V,E	V	-
Sustainability officer	V	V	V	-	V	V	V,A	-
System administrator	-	V	-	-	-	-	-	C,V,E

C = create; V = view; E = edit permitted fields or record an operational confirmation; A = decide an assigned approval or confirmation stage; - = no access through that role alone.

How to read the matrix. Each permission is limited to the user's department, assigned work or linked request. Viewing a published listing gives access to its marketplace summary, not the owning department's private asset documents. Requesters can view the listed asset summary and their own linked transactions. All nine roles may search authorised published listings; the System administrator's V under Listings is this ordinary staff access, not permission to edit business records. A user who needs another function must also hold its required role.

Approval limits. A is permission to decide only the stage assigned in Table 5.2. Asset confirmation is by an Asset custodian or Department representative. Listing release and request approval are by the relevant Department representative. A reservation hold is confirmed by the releasing Asset custodian or Department representative. Transfers require both department representatives, followed by Finance and/or Procurement only when BR-06 applies. Repair spending requires the owning Department representative and Finance only when BR-07 applies. End-of-life review is by Sustainability, then Finance write-off approval. BR-02 blocks the creator or requester from deciding their own approval task even if they hold several roles; an eligible alternate is used under TBD-9.

Editing limits. E does not authorise changing an already approved decision or editing an audit entry. A Requester can edit a draft, revise a request with renewed approval where required, or cancel a permitted request or reservation. Custodians record verification, handover and receipt; Maintenance staff record assessment and work results. Department representatives may maintain their department's listings and cancel permitted reservation holds. End-of-life C and E allow the department or custodian to prepare a proposal and record completion evidence after approval. Approved quantity, value, recipient or action changes require a new decision under FR-10.7.

Administration and reports. Business administrator C,V,A under Users and roles covers requesting and authorising role assignments; System administrator C,V,E covers account maintenance and

applying assignments already authorised under BR-01. Neither receives transfer, spending or disposal approval through technical or business administration alone. For reports, the Business administrator reviews operational figures, Finance reviews cost and savings figures, and Sustainability reviews sustainability figures (BR-10); Procurement can inspect relevant reports but is not a substitute for those data owners.

2.4 Operating Environment

Dawra Campus shall provide an English web interface for laptops and phones. The proposed environment below follows the technology choices from Phase 2. These choices still need testing before they are confirmed. The classic features, such as asset registration, search and approval workflows, shall run without a paid AI account or an AI API key, as required by NFR-09.

Component	Environment	Status
User devices and browsers	Existing student or university computers running Windows, macOS or Linux, and phones running Android or iOS. Users access the system through a browser, such as Chrome, Edge, Firefox or Safari. No separate phone application is required.	Supported browser and operating-system combinations will be confirmed through testing. The tested versions will be recorded.
Application server	Python with Django 5.2 LTS. Django will produce the web pages and manage forms, accounts and workflows. A separate AI adapter will connect the application to AI services.	Selected candidate from Phase 2. We will test the application and record the Python version and patched Django release used.
Database	PostgreSQL will store accounts, departments, assets, requests, approvals, transfers and other project records. The selected PostgreSQL version must work with the chosen Django release.	Candidate database. Its version, connections and data recovery process still need testing.
File storage	Photos and documents will use persistent storage, meaning that uploaded files remain available after the application restarts or is redeployed. Access to files will follow the user's permissions.	Storage provider still to be selected. File access, recovery and retention after a restart or redeployment must be tested.
Hosted AI services	The Phase 2 candidates are text-embedding-3-small for matching requests to assets and gpt-4.1-mini-2025-04-14 for category suggestions, the assistant, explanations and report text. Requests will pass through the AI adapter.	Approved access, cost, data handling and output quality still need checking. These services are not required to start the classic system.
Local AI alternative	A MiniLM embedding model may be used locally for matching. It does not replace the model used for the assistant or generated reports.	Alternative candidate. Memory use and matching quality must be tested before adoption.
Local development and grading	The application and database shall run on an existing student or grader's computer using the repository's setup instructions, configuration examples and synthetic test data. Classic features shall work without AI credentials.	Setup instructions and a fresh-install test are required under NFR-09. Hardware needs will be checked during the prototype trial.
Hosting	Render is the proposed application host, with Neon providing the PostgreSQL database. Uploaded photos and documents require separate persistent storage.	Proposed hosting arrangement from Phase 2. Deployment, access controls, stored data and recovery must be tested. The first request after inactivity will be measured separately from normal use.

The setup guide shall record the tested operating systems, browsers, hardware and software versions so another team member or grader can repeat the setup. Any change to the proposed technologies shall include a recorded reason, as required by C-01.

The prototype shall follow the AED 0 budget in C-02. Any service that introduces a cost requires instructor approval. If an AI service is unavailable or has not been configured, users shall be able to continue through the classic features, as required by BR-15 and NFR-09.

2.5 Design and Implementation Constraints

ID	Constraint
C-01	We will use the tools picked in Phase 2 (Django 5.2 LTS, PostgreSQL and a separate AI adapter), unless testing in Phase 4 shows they won't work. If we change, we will write down why.
C-02	Our budget is AED 0, so we only use free tiers, university services or open-source tools unless the instructors approve a cost.
C-03	We only store made-up or anonymised data, never real KU records or staff details.
C-04	There is no live connection to KU's ERP, finance, procurement or login systems.
C-05	AI features use existing (pretrained) models through the AI adapter; we don't train our own.
C-06	Only the fields an AI task needs are sent to an AI service, never passwords or unnecessary personal data.
C-07	The classic system has to install and run from the repository without a paid account or an AI key (NFR-09).
C-08	All work goes through our GitHub repository, with each member committing their own work and teammates reviewing pull requests.
C-09	The system is a website in English that works on laptops and phones. No phone apps and no Arabic version in this release.
C-10	The course sets the schedule: design in Phases 4–5 (week of 26 October), first build in Phase 6 (week of 16 November) and the final demo in Phase 8.

2.6 User Documentation

ID	Document	Who it's for
D-01	Setup guide (README)	Graders and developers
D-02	One-page guides for requesters, custodians, department representatives and approvers	Staff using the system
D-03	Help text next to forms and status labels	All users
D-04	Admin guide for roles, departments, categories, approval settings and AI settings	Administrators
D-05	A "How AI suggestions work" page explaining scores, confidence and how to override them	All users
D-06	A short note on how each sustainability number is calculated (section 6.1)	Sustainability and finance staff

2.7 Assumptions and Dependencies

Assumptions

- A-01: The lab instructors are our client and will confirm the requirements and the TBD values in Appendix C.

- A-02: Made-up data of the size planned in Phase 1 (500 assets, 10 departments, 50 requests, 20 labelled matching cases) is enough to show and test every feature.
- A-03: The nine roles from Phase 1 cover everyone involved, and one person can have more than one role.
- A-04: One asset record can be a batch of identical items with a quantity, for example 20 identical chairs.
- A-05: The current KU process in Phases 1 and 2 is our best reconstruction; we haven't checked it with KU staff.

Dependencies

- DEP-01: The AI features need access to the AI services we picked, at an acceptable cost (Phase 2, section 7). Without it we use the local model or the classic features.
- DEP-02: Putting the site online depends on the free hosting tier from Phase 2, plus separate storage for uploaded files.
- DEP-03: Carbon estimates need published emission factors for each category (TBD-6).
- DEP-04: Approval limits and special-handling rules need KU's real values (TBD-1, TBD-2, TBD-8).

3. External Interface Requirements

3.1 User Interfaces

Dawra Campus shall provide a responsive English website usable in laptop and phone browsers. Screens shall share the same navigation, form conventions and status labels while showing only actions allowed by the user's active role and department. Users shall be able to see which department they are working in and switch among their authorised assignments. This section defines what the screens must do; detailed visual mockups belong to the design phase.

ID	Screen	What it must show and allow	Main users
UI-01	Sign in	The system shall provide the sign-in form, clear validation messages and an account-status or temporary-lock message where applicable. After successful sign-in, it shall open the user's authorised dashboard.	All nine staff roles
UI-02	Role dashboard	The system shall show the active role and department, pending and overdue tasks, own requests, reservations close to expiry, assets needing verification and unread notifications. Each task shall open its authorised record.	All nine staff roles
UI-03	Asset registration	The system shall show the fields in FR-04.1, identify required fields, validate quantities and uploads, warn about possible duplicates and provide draft and confirmation actions. Its AI panel shall let the user accept, edit or reject category and description suggestions; manual entry shall remain available.	Asset custodian, Department representative

ID	Screen	What it must show and allow	Main users
UI-04	Asset record and history	The system shall show authorised asset details, condition, life-cycle state, verification date, available/reserved/allocated quantities, custody, evidence and chronological history. It shall show editing or confirmation controls only to the roles allowed by section 2.3.	Asset custodian, Department representative; other roles with access to the record
UI-05	Marketplace search	The system shall provide keyword search, combined filters and sorting from FR-08. Results shall show listing quantity, condition, location, restrictions and verification date. It shall distinguish no results from a service error, offer a request form when appropriate, and keep classic search available without AI.	All staff users, especially Requester
UI-06	Resource request	The system shall collect category, purpose, specifications, quantity, minimum condition, urgency, delivery location, required-by date and optional description. It shall show draft, submission, cancellation and tracking controls appropriate to the request state. Submission shall identify the department approval still required.	Requester; Department representative reviews through UI-08
UI-07	Match review	The system shall show only actionable matches that pass BR-14. For each match it shall show compatibility score, reasons, missing information and separate confidence information or its unavailability. The user shall be able to inspect another eligible match and continue to reservation. Selecting a match shall not itself reserve or approve it.	Requester; Asset custodian or Department representative inspecting a linked request
UI-08	Approvals inbox	The system shall show each assigned stage, parent-record state, creator, quantity, cost, evidence, AI explanation where available and the version awaiting decision. It shall provide approve, reject and request-changes controls only for the assigned role, enforce BR-02 and require a reason for rejection or requested changes.	Department representative, Finance officer, Procurement officer, Sustainability officer; Business administrator for role authorisation and permitted task routing
UI-09	Transfer handover and receipt	The system shall show both departments and custodians, required approvals, quantity, arrangements and planned date. It shall enable handover after TR-04 and receipt after TR-05. It shall collect actual condition and quantity, provide a discrepancy action and show that ownership changes only after receipt under BR-13.	Releasing and receiving Asset custodians; Department representatives view decisions and disputes
UI-10	Maintenance work order	The system shall show the asset and inspection history, defect, assessment, estimated cost, approval status, assigned work, parts, labour, actual cost and outcome. Work and completion controls shall follow MW states. Increased costs shall return for approval before completion.	Asset custodian, Maintenance staff, Department representative; Finance officer where BR-07 applies

ID	Screen	What it must show and allow	Main users
UI-11	End-of-life review	The system shall show the proposed action, quantity, reasons, evidence and the reuse-first alternatives. It shall show an optional sustainable-action recommendation, special-handling/EHS and data-erasure evidence where required, and separate Sustainability and Finance decisions. Completion evidence shall be recorded only after approval.	Department representative, Asset custodian, Sustainability officer, Finance officer
UI-12	Campus assistant	Where the desirable assistant feature is available, the system shall show authorised answers, supporting records where available, uncertainty and any proposed field changes. It shall require user confirmation before passing an action to the normal form and approval workflow. If unavailable, it shall direct the user to classic search, forms or help.	All nine staff roles
UI-13	Reports and sustainability dashboard	The system shall show report type, selected period and authorised departments, totals, units, source records and limitations. It shall distinguish measured, estimated and synthetic values and show N/A for missing inputs. Conventional reports shall remain available; generated narrative shall be labelled Draft until the correct data owner reviews it.	Business administrator, Finance officer, Procurement officer, Sustainability officer
UI-14	Administration	The system shall provide accounts and authorised role-application controls for System administrators and business-setting/role-authorisation controls for Business administrators. It shall show departments, categories, control flags and current proposed thresholds. Where configurable settings are implemented, it shall record changes and effective dates under FR-03.3.	Business administrator, System administrator

3.1.1 Interface Standards

- The system shall use the exact state names in section 4.2, such as Pending department approval, Published and Disputed, and show the next responsible role where an action is outstanding. Colour alone shall not communicate status.
- The system shall show the user's active department and role, consistent navigation, a page title and the unread-notification count. On narrow screens, tables shall remain readable through wrapping or controlled horizontal scrolling.
- Before an approval, rejection, withdrawal, cancellation, handover, receipt or end-of-life completion, the system shall show a confirmation containing the affected record and consequential changes. Rejections, requests for changes and cancellations shall require a reason under BR-20.
- The system shall label generated content as AI suggestion or AI draft, show confidence separately from compatibility, and allow acceptance, correction or rejection where the workflow permits. Unavailable confidence or model-version information shall be marked as unavailable.

- The system shall show an actionable explanation beside invalid fields, retain entered information after validation or service errors, and keep the form available for correction. Permission errors shall not reveal restricted records or credentials.
- The system shall check authorisation on the server as well as controlling visible buttons. A hidden button shall not be treated as the security control, and a confirmed assistant action shall follow the same checks as a classic form (BR-01, BR-15).
- While a submission is being processed, the system shall show its progress and prevent repeated activation from creating duplicate records. It shall recheck the current quantity and approved record version before applying a consequential change.
- The system shall use English labels and help text, visible field labels, plain-language instructions and consistent date formatting as DD Mon YYYY. Money shall show its currency, normally AED; quantities and sustainability figures shall show their units. Audit timestamps shall show the applicable time zone.
- The interface shall target WCAG 2.2 AA, as proposed in the template, with keyboard-operable controls, visible focus, sufficient contrast, meaningful image descriptions and errors associated with their fields. Accessibility shall be verified during testing rather than claimed as already achieved.
- If an AI call fails, times out or returns unusable output, then the system shall explain the issue, preserve entered data and provide the relevant classic alternative under AI-13 and NFR-07. It shall never show a generated answer as an approval.
- The system shall show the proposed values identified in Appendix C as proposed, and identify optional or unavailable features without preventing completion of the essential classic workflow.

3.2 Hardware Interfaces

The system shall work through a browser on ordinary laptops, desktop computers and phones. Users shall be able to upload asset photographs taken with a phone camera, subject to the file limits in FR-04.1. The prototype shall not require dedicated hardware; barcode and RFID scanners are outside its scope.

3.3 Software Interfaces

The following interfaces connect Dawra Campus to other software and storage services. The proposed technologies are listed in section 2.4. Their exact versions and settings will be recorded in the setup guide after testing.

ID	Interface	Data exchanged	Purpose
SI-01	PostgreSQL database through Django	The application reads and writes accounts, departments, assets, requests, reservations, approvals, transfers, maintenance records, AI output records and audit entries. The database returns the requested records or the result of an update.	Store and retrieve project records while preserving the access rules and data integrity requirements.

ID	Interface	Data exchanged	Purpose
SI-02	AI adapter and the models listed in section 2.4	The adapter sends only approved fields needed for the task, such as asset descriptions, categories, specifications, condition, quantity and request details. It processes model responses and returns suggestions, matching scores where relevant, explanations, confidence information or its unavailability, and the model identifier and version.	Keep AI connections separate from the main application and check responses before presenting them to users.
SI-03	Persistent file storage; provider to be confirmed	The application sends uploaded photographs and supporting documents, together with their record references. Storage returns a file reference and allows authorised retrieval. Uploads must meet the file type, size and quantity limits in FR-04.1.	Keep supporting evidence available after restarts and redeployments while restricting access to authorised users.
SI-04	Export and printable views	Transfer exports contain the asset ID, quantity, ownership and custody details, locations, approvals and handover and receipt confirmations. Audit exports contain the selected audit entries. Report exports contain the selected filters, records and totals.	Support manual checking against the existing KU asset register under FR-11.7 and audit export under FR-18.3. CSV and printable report export remains Desirable under FR-17.3.

The system shall apply the user's access permissions before retrieving data for AI processing, displaying files or producing exports. Exported records support manual reconciliation; the prototype shall not connect directly to KU's asset register, finance, procurement or identity systems, as stated in C-04.

If the AI service is unavailable, times out or returns an invalid response, the system shall show a clear message and provide the relevant classic search, form or report. It shall preserve the user's input and keep all normal permission and approval checks in place, as required by AI-13, NFR-07 and BR-15.

3.4 Communications Interfaces

- The system shall use HTTPS for browser communication to protect data while it is being sent.
- The AI adapter shall communicate with external AI services through HTTPS using JSON, a structured format for exchanging data. It shall check that each response contains the fields required for its task before using it.
- If an AI request fails or remains unanswered, the system shall display a failure or timeout message and make the relevant classic alternative available within 15 seconds of the original request. Any retries shall remain within that time limit. User input shall be preserved, as required by NFR-07.
- The system shall keep AI service credentials on the server and use them only to authenticate service calls. It shall exclude user passwords, service credentials and unnecessary personal data from AI input content and ordinary logs.

- The system shall provide notifications inside the application and show them only to users who may access the related record, as required by FR-16. Email delivery is a Future enhancement under FR-16.6 and is not required for the prototype.

4. System Features

Section 4.1 shows the use-case model, section 4.2 defines the states that records move through, and sections 4.3 to 4.23 specify the requirements feature by feature.

4.1 Use-Case Model

4.1.1 Actors

The nine roles in section 2.3 are primary actors. Every role can sign in, view its authorised dashboard and notifications, search permitted published listings, and use the desirable assistant when available, while retaining its own department and record restrictions. The diagram shows the roles that initiate each use case; other participants and approval stages are explained in the descriptions below. One person can perform several roles, but BR-02 still prevents self-approval. The external AI service is a supporting actor; it returns suggestions and never authorises a business action. KU EHS consultation is recorded evidence in the end-of-life workflow, not a tenth application role.

Actor	Type	Goal in the system
Department representative	Primary	Confirm records and decide the assigned department stages for listings, requests, transfers and spending.
Asset custodian	Primary	Maintain assets and listings, confirm holds, record handover/receipt and support maintenance and retirement records.
Requester	Primary	Find an eligible resource, obtain request approval and seek a reservation and transfer.
Business administrator	Primary	Authorise role assignments, maintain business settings, route exceptions and review operational reports and audit records.
Procurement officer	Primary	Decide procurement-controlled transfer stages and inspect relevant policy and reuse reports.
Finance officer	Primary	Decide required financial stages and review financial reports and write-offs.
Maintenance staff	Primary	Inspect assigned assets and complete approved maintenance with recorded costs and outcomes.
Sustainability officer	Primary	Review reuse-first end-of-life decisions and verify sustainability reporting.
System administrator	Primary	Maintain accounts and apply authorised roles while preserving the business-approval boundary.
AI service	Secondary; external system	Return category suggestions, eligible-match rankings, optional action advice and assistant answers, and draft report text when requested.

4.1.2 Use-Case Diagram

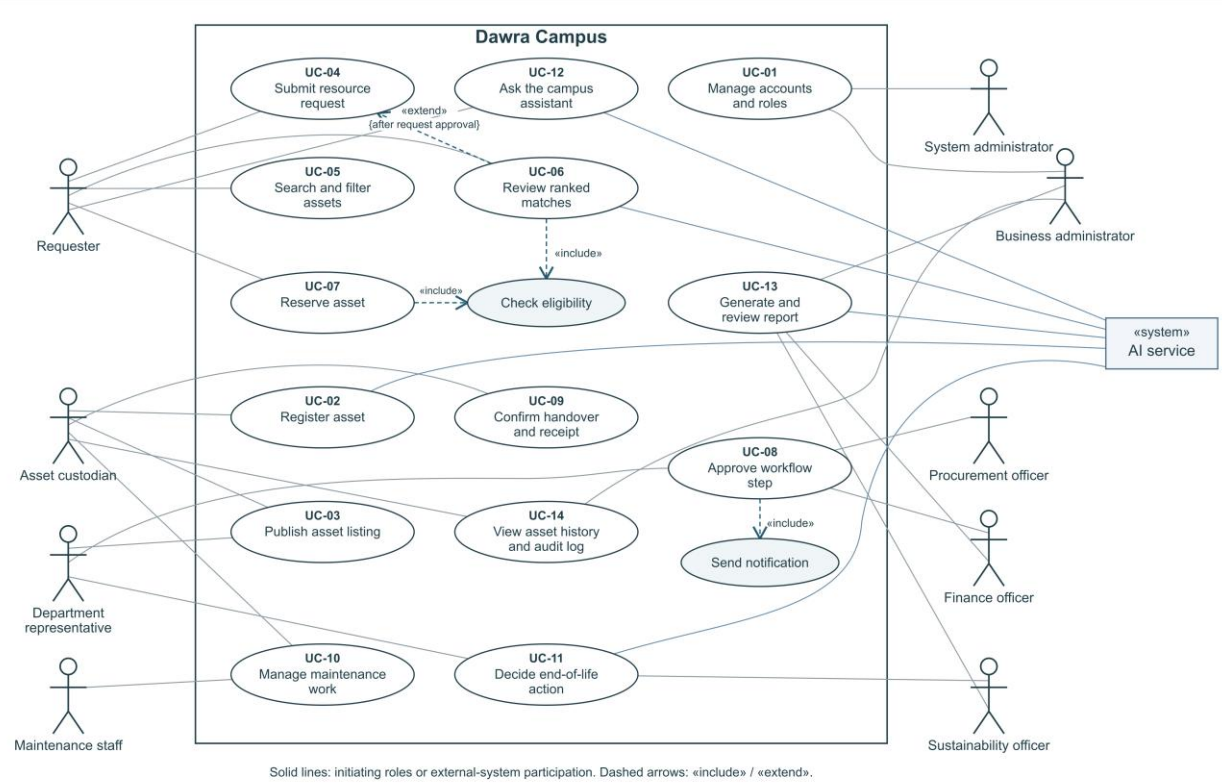


Figure 1. Dawra Campus use-case diagram

Every role can also search (UC-05) and use the assistant (UC-12); to keep the diagram readable these are shown for the Requester only.

The rectangle is the Dawra Campus system boundary; staff actors and the external AI service are outside it. Solid lines show initiating roles or external-system participation, not unrestricted permission. Approval-stage participation is represented through UC-08 rather than extra connections to each workflow. Other participants remain documented in the use-case descriptions. Check eligibility is always included when ranking or reserving; Send notification is included when an approval decision changes the workflow. Review ranked matches optionally extends an approved resource request. AI classification can fall back to manual entry, and sustainable-action advice is desirable, so neither is modelled as a mandatory AI dependency.

4.1.3 Use-Case Descriptions

ID	Use case	Main actors	Related requirements
UC-01	Manage accounts and roles	Business administrator, System administrator	4.3 (FR-01, FR-02), BR-01, BR-02
UC-02	Register asset	Asset custodian, Department representative, AI service	4.5 (FR-04, FR-05), 4.17 (AI-01)
UC-03	Publish asset listing	Department representative, Asset custodian	4.6 (FR-06), BR-04
UC-04	Submit resource request	Requester, Department representative	4.7 (FR-07), BR-03
UC-05	Search and filter assets	Requester, all staff users	4.8 (FR-08)
UC-06	Review ranked matches	Requester, AI service	4.18–4.19 (AI-02 to AI-05)
UC-07	Reserve asset	Requester, Asset custodian	4.9 (FR-09), BR-11, BR-12
UC-08	Approve workflow step	Department representative, Finance officer, Procurement officer, Sustainability officer; Business administrator and Asset custodian for their assigned stages	4.10 (FR-10), BR-01 to BR-10
UC-09	Confirm handover and receipt	Asset custodian, Department representative	4.11 (FR-11), BR-13
UC-10	Manage maintenance work	Asset custodian, Maintenance staff, Department representative; Finance officer when required	4.12 (FR-12, FR-13), BR-07
UC-11	Decide end-of-life action	Sustainability officer, Finance officer, Department representative, Asset custodian, AI service	4.13 (FR-14), 4.20 (AI-06), BR-08, BR-09
UC-12	Ask the campus assistant	All staff users, AI service	4.21 (AI-07, AI-08)
UC-13	Generate and review report	Business administrator, Finance officer, Procurement officer, Sustainability officer, AI service	4.16 (FR-17), 4.22 (AI-09, AI-10), 6.1
UC-14	View asset history and audit log	Asset custodian, Business administrator	4.14 (FR-15, FR-18)

The descriptions below define proposed prototype behaviour using the existing requirement IDs and states. They do not report an implemented or observed KU workflow. Approval owners are defined in Table 5.2; thresholds and special-handling rules retain their Appendix C status. Each alternative is numbered after the main step from which it branches.

UC-01 Manage accounts and roles

Actors: System administrator, Business administrator; affected Staff user.

Related requirements: FR-01.1 to FR-01.4, FR-02.1 to FR-02.4, FR-03.1 to FR-03.5; BR-01, BR-02.

Precondition: The administrators are signed in with the relevant assignments. Any department and role being assigned exist in the configuration.

Main success scenario:

1. The System administrator opens UI-14 and creates or updates an account with its name, staff email, department and account status.
2. A Business administrator requests the required role assignments for named departments.
3. The system checks the assignment and presents it to an eligible Business administrator for authorisation. The creator cannot decide their own approval task (BR-02).
4. The authorised Business administrator confirms the assignment. The system records the decision and makes it available for application.
5. The System administrator applies the authorised roles. The system records who authorised and applied them without adding unrelated business permissions.
6. The staff user signs in through UI-01. The system checks the active account and shows UI-02 with only permitted functions.

Alternative and exception flows:

1a. An account field is invalid or already identifies another account: the system explains the issue and saves no conflicting account.

1b. The Business administrator chooses business settings in UI-14: they maintain departments, categories and control flags under FR-03. Where configurable settings are available, changes record an effective date without changing approvals in progress. Referenced departments or categories can be deactivated instead of deleted.

3a. The authoriser is the creator or lacks the required role: the system blocks the decision and routes it to an eligible alternate under BR-02 and TBD-9.

4a. The assignment is rejected: the system closes the assignment request with a reason and does not activate the role.

5a. The System administrator attempts an unauthorised assignment: the system refuses and logs the attempt.

6a. The account is deactivated: the system refuses sign-in, ends active sessions and lists open approval tasks for reassignment. Five consecutive failed sign-in attempts trigger the lock in FR-02.4.

Postcondition: The account and approved roles are recorded and active where authorised, with an audit trail. Failed or rejected assignments grant no additional access. Business settings under FR-03 remain restricted to the Business administrator.

UC-02 Register asset

Actors: Asset custodian, Department representative; AI service supports classification when available.

Related requirements: FR-04.1 to FR-04.6, FR-05.1 to FR-05.4; AI-01, AI-12, AI-13; BR-02, BR-11, BR-15.

Precondition: The user is signed in and may maintain assets for the owning department. The department and relevant categories exist.

Main success scenario:

1. The Asset custodian or Department representative opens UI-03 and enters the fields in FR-04.1, including quantity, unit, ownership, custody, condition, specifications and available evidence.
2. The system assigns an asset ID, checks required fields and upload limits, warns about possible duplicates and keeps the unconfirmed record in Draft (AS-01).
3. When AI is available, the system requests classification and description suggestions and labels the response as AI suggestion, with provenance and confidence information where available.
4. The user accepts, edits or rejects the suggestions and checks the entered information.
5. An eligible Asset custodian or Department representative who did not create or request the record confirms it under Table 5.2 and BR-02.
6. The system saves the confirmed asset as In service (AS-02) or In storage (AS-03), records the confirming user and verification date, and keeps the available, reserved and allocated quantities consistent.

Alternative and exception flows:

2a. Required information or an upload is invalid: the system retains Draft and identifies the fields or files to correct.

2b. A possible duplicate exists: the user reviews the existing record and explicitly confirms a distinct asset before saving.

3a. AI is unavailable or produces an unusable suggestion: the system preserves the form and allows manual category and description entry; no AI result is invented.

5a. Confirmation is rejected or the user cannot decide the task: the record remains Draft and is sent to an eligible confirmer where required.

Postcondition: A valid confirmed asset is available to its authorised department users, with quantities, confirmation and history recorded. A draft asset cannot be published or reserved.

UC-03 Publish asset listing

Actors: Asset custodian, Department representative of the owning department.

Related requirements: FR-06.1 to FR-06.6, FR-05.4, FR-10.1 to FR-10.4; BR-02, BR-04, BR-16, BR-19, BR-20.

Precondition: The asset is confirmed, is In service or In storage, and has an eligible quantity the owning department can offer.

Main success scenario:

1. The Asset custodian or Department representative prepares a Draft listing (LS-01) with offered quantity, condition, location, available-from date, restrictions and reason for release.
2. The system validates the quantity and asset state and records the listing when the user submits it.
3. The system sets Pending release approval (LS-02) and notifies an eligible owning-department representative through UI-08.
4. The representative reviews the listing and independently approves release under BR-04 and BR-02.
5. The system sets Published (LS-03), shows the listing to authorised staff through UI-05 and notifies the custodian.

Alternative and exception flows:

2a. The quantity is invalid or the asset is Under maintenance, In transfer or Retired: the system blocks submission and explains the restriction.

4a. Changes are requested or release is rejected: the listing returns to Draft with the decision reason and remains hidden from other departments.

5a. Verification is older than TBD-4: the published listing shows Verification needed, and reservation confirmation is blocked until the custodian verifies it again.

5b. The owning department withdraws the listing: it becomes Withdrawn (LS-05), pending reservations are cancelled and requesters are notified. BR-19 blocks withdrawal during an approved transfer.

Postcondition: Only an approved listing is visible for reuse. Its quantity, restrictions and verification status are shown, while release evidence and decisions remain traceable.

UC-04 Submit resource request

Actors: Requester (initiator), receiving Department representative.

Related requirements: FR-07.1 to FR-07.8, FR-10.1 to FR-10.4; BR-02, BR-03, BR-18, BR-20.

Precondition: The Requester is signed in with an authorised department. The required category exists; an assistant draft, if used, has not yet submitted the request.

Main success scenario:

1. The Requester opens UI-06 and enters the purpose, category, specifications, quantity, minimum condition, urgency, delivery location, required-by date and optional description.
2. The system validates the information and lets the Requester review the draft (RQ-01).
3. The Requester confirms submission. The system sets Pending department approval (RQ-02) and creates the department approval task.
4. An eligible receiving Department representative reviews the need and approves it through UC-08.
5. The system sets Open (RQ-03), notifies the Requester and shows the next available actions: classic search, optional ranked-match review and reservation of an eligible listing.

Alternative and exception flows:

2a. The date is in the past or the quantity is not a positive whole number: the system preserves the form and explains the correction.

4a. Changes are requested: the request returns to Draft with the reason. A rejection sets Rejected (RQ-08) and records the reason.

5a. The Requester changes approved quantity, specifications or date: the system returns the request to Pending department approval before further fulfilment.

5b. The Requester cancels before fulfilment: the request becomes Cancelled (RQ-09) and any linked hold is released as required by BR-20.

5c. No eligible resource exists: the Requester may close the request as No suitable item (RQ-07); purchasing remains outside the system and no avoided purchase is counted.

5d. The required-by date passes before fulfilment: the system sets Expired (RQ-10), releases a linked reservation and notifies the Requester.

Postcondition: A submitted request has a traceable department decision. Matching and reservation require department approval; fulfilment requires an approved transfer and confirmed receipt through UC-09. Submission or an AI suggestion alone grants no approval.

UC-05 Search and filter assets

Actors: Staff user, especially Requester.

Related requirements: FR-08.1 to FR-08.6, FR-02.2, FR-06.3, FR-06.4; AI-13; BR-14, BR-16.

Precondition: The user is signed in. The system knows which published summaries and department records the user may view.

Main success scenario:

1. The user opens UI-05, enters keywords and selects filters such as category, condition, department, location and available quantity.
2. The system searches authorised records using the combined criteria and returns the result count and permitted sort options.
3. The user opens a listing to inspect its specifications, condition, quantity, restrictions and last verification date.
4. The user refines the criteria or chooses a listing for an Open request. The reservation workflow rechecks all mandatory constraints before confirming a hold.

Alternative and exception flows:

2a. No listing matches: the system explains the empty result and offers to start a request using the same criteria.

2b. AI is unavailable: classic keyword search, filters and sorting still work.

3a. The listing is unpublished: another department cannot view it. An expired verification displays Verification needed and prevents an unverified reservation confirmation.

4a. Availability changes after the results were displayed: the system shows the current quantity and requires a fresh eligibility check instead of relying on an old search result.

Postcondition: The user has inspected authorised results. Searching or selecting a result does not create a reservation, approval or ownership change.

UC-06 Review ranked matches

Actors: Requester; AI service (supporting).

Related requirements: AI-02 to AI-05, AI-11 to AI-13; FR-08.6, FR-09.1; BR-14, BR-15.

Precondition: The Requester has an Open request and may view its candidate listings. Ranked matching is available, or a classic alternative can be used.

Main success scenario:

1. The Requester opens UI-07 from an approved request and asks to review matching resources.
2. The system performs Check eligibility using access rights, availability, quantity, required condition, compatibility and required date before sending eligible candidates for AI ranking.
3. The AI service ranks the eligible candidates. The system records the task, model identifier/version where available and response time.
4. The system displays ranked matches with compatibility scores, explanations, unresolved information and separate confidence information or its unavailability.
5. The Requester reviews the evidence and selects a preferred eligible result, including another eligible result instead of the first-ranked one. The system records the user's response to the recommendation.
6. The system offers UC-07 for the selected listing and rechecks eligibility when a hold is requested.

Alternative and exception flows:

2a. No eligible candidate remains: the system gives an empty-result explanation and offers classic search or the No suitable item outcome in UC-04.

3a. AI fails, times out or returns invalid or insufficiently confident output: the system preserves the request and makes classic search and filters available under AI-13.

4a. Evidence is incomplete: the system identifies missing information and does not make a candidate actionable unless the mandatory constraints can be confirmed.

5a. The user rejects all suggestions: no quantity is held; the user may refine the request or continue with classic search.

Postcondition: The reviewed recommendation and user response are recorded. Selecting a result does not reserve it or bypass approval. This use case optionally extends UC-04 after department approval.

UC-07 Reserve asset

Actors: Requester, releasing Asset custodian or Department representative.

Related requirements: FR-09.1 to FR-09.7, FR-11.1, FR-06.4, FR-05.3; BR-02, BR-11, BR-12, BR-14 to BR-16, BR-20.

Precondition: The Requester has an Open request (RQ-03) and has selected a Published listing (LS-03) with a suitable quantity.

Main success scenario:

1. The Requester chooses the listing and quantity and confirms a reservation request.
2. The system checks permissions, quantity and mandatory constraints, creates Requested (RV-01) and sets the request to Reservation requested (RQ-04).
3. The releasing Asset custodian or Department representative checks the stock and confirms current condition and availability. If verification is stale, the custodian verifies the asset before confirmation.
4. An eligible releasing user confirms the hold independently of the requester under Table 5.2.
5. The system records the hold and updates the available quantity together, preventing another user from receiving the same units, and sets Confirmed (RV-02). A listing with no remaining offered quantity becomes Fully reserved (LS-04).
6. The system shows the hold expiry from TBD-3 and notifies both sides that transfer approval is still required.

Alternative and exception flows:

2a. The requested quantity is invalid or unavailable: the system refuses it and shows the current quantity.

4a. The releasing side declines: the reservation becomes Declined (RV-03), a reason is recorded and the request returns to Open.

5a. Two users attempt to hold the last units simultaneously: only one confirmation succeeds; the other is told that the quantity is no longer available.

6a. The hold expires without an approved transfer: the system sets Expired (RV-05), releases the quantity and notifies both sides. A permitted cancellation uses RV-06 and also releases the hold.

6b. The Requester proceeds with the confirmed hold: the system creates a transfer in Pending release approval (TR-01), links the reservation as Converted to transfer (RV-04) and sets the request to In transfer (RQ-05). UC-08 carries out the required approvals before UC-09.

Postcondition: A successful reservation holds the quantity for the configured period; it is not transfer approval. Rejected, expired or cancelled holds leave no duplicate allocation.

UC-08 Approve workflow step

Actors: Department representative, Finance officer, Procurement officer, Sustainability officer; Business administrator for role authorisation; Asset custodian for record/hold confirmations in Table 5.2.

Related requirements: FR-10.1 to FR-10.8, FR-16.1 to FR-16.5; BR-01 to BR-10, BR-20.

Precondition: A Pending task (AT-01) exists for an allowed stage in Table 5.2. The user has the assigned role and department and is not the requester or creator of that approval record.

Main success scenario:

1. The approver opens the assigned task in UI-08.
2. The system shows the record version, creator, relevant asset and quantity, costs, evidence, prior decisions and any AI suggestion.
3. The system checks that the user may decide this stage and that earlier required stages have been completed.
4. The approver reviews the evidence and confirms Approve.
5. The system records Approved (AT-02) with the user, role, time and approved record version, and advances the parent record according to section 4.2 and Table 5.2.
6. The system performs Send notification to the users involved and creates the next required task where applicable. A transfer reaches Approved (TR-04) only after both department stages and any required policy stages.

Alternative and exception flows:

3a. The user lacks the assigned role or is the creator/requester: the system refuses the decision. The Business administrator may reassign it only to an eligible user.

4a. The approver rejects or requests changes: the system requires a reason, saves AT-03 or AT-04 and uses the parent's defined rejection or revision path. Held quantity is released where the rejected workflow requires it.

5a. The record version or decision has changed: the system refuses a stale or duplicate decision and shows the current task. Changes affecting an existing approval create a new task under FR-10.7.

6a. The task stays pending beyond TBD-5: the system marks it overdue and reminds the assigned approvers; overdue status does not automatically approve it.

Postcondition: One authorised decision is recorded and the parent follows its allowed transition. No earlier stage, required approver or human decision is bypassed.

UC-09 Confirm handover and receipt

Actors: Releasing and receiving Asset custodians; Department representatives inspect decisions and resolve disputes.

Related requirements: FR-11.1 to FR-11.7, FR-05.3, FR-05.4; BR-05, BR-06, BR-11, BR-13, BR-20.

Precondition: The linked transfer is Approved (TR-04). Both department approvals and any Finance/Procurement stage required by BR-06 are complete; the agreed quantity remains allocated to this transfer.

Main success scenario:

1. The releasing and receiving custodians open UI-09 and confirm the collection or delivery arrangement and planned date.
2. The releasing custodian records actual quantity, condition and handover evidence and confirms handover.
3. The system sets Handed over (TR-05) and the asset to In transfer (AS-05). Ownership remains with the releasing department while receipt is outstanding.
4. The receiving custodian checks the delivered quantity and condition against the agreed transfer and confirms receipt.
5. The system sets Completed (TR-06), updates the owning department, custodian and location under BR-13, moves the received asset to In service or In storage, and sets the linked request to Fulfilled (RQ-06).
6. The system records both confirmations in the history, notifies the parties and provides the transfer record for reconciliation with the existing register.

Alternative and exception flows:

1a. A required approval is missing: the system blocks handover and identifies the outstanding stage.

2a. The transfer is cancelled before handover: the system sets Cancelled (TR-09), releases the held quantity and records the reason.

4a. Quantity or condition differs: the system sets Disputed (TR-07), retains the existing ownership and notifies both department representatives. Resolution follows the TR-07 paths: receipt and completion, or return and cancellation.

5a. Another user has already confirmed receipt: the system displays the completed record and does not apply ownership or quantity changes twice.

Postcondition: Ownership, custody and location change only after a valid receipt. An unresolved dispute leaves ownership unchanged and remains visible to the responsible users.

UC-10 Manage maintenance work

Actors: Asset custodian, Maintenance staff, owning Department representative; Finance officer where the spending threshold applies.

Related requirements: FR-12.1, FR-12.2, FR-13.1 to FR-13.7; BR-02, BR-07, BR-20.

Precondition: The asset exists and the user may report or maintain it. Approval thresholds retain their Appendix C status.

Main success scenario:

1. The custodian or department representative records a defect through UI-10. The system opens Reported (MW-01) and notifies Maintenance staff.
2. Maintenance staff inspect the asset, record findings, condition and evidence, and assess the work and estimated cost in Under assessment (MW-02).
3. The system sends the estimate to Awaiting cost approval (MW-03). The owning Department representative decides first; Finance also decides when BR-07 applies.
4. After the required approvals, the system sets Approved (MW-04) and keeps the asset Under maintenance (AS-04), unavailable for publication or reservation.
5. Maintenance staff carry out the approved work in In progress (MW-05) and record parts, labour, actual cost, outcome and resulting condition.
6. The system checks spending approval before setting Completed (MW-06), restores the asset to its previous In service or In storage state, and records the result in its history.

Alternative and exception flows:

2a. Inspection identifies an unsafe or Not working item: publication and reservation are blocked under FR-12.2.

3a. Spending is rejected: the system sets Rejected (MW-08), records the reason and allows no unapproved work.

5a. Actual cost would exceed the approved amount: the work order returns for cost approval under BR-07; it cannot be completed until the difference is approved.

5b. The item is Not repairable: the system records MW-07, sets the asset to Retired (AS-06) and opens an end-of-life proposal (EL-01) for UC-11.

5c. A permitted withdrawal cancels the work order: the system records MW-09 with a reason and follows the allowed asset transition without treating unfinished work as a completed repair.

Postcondition: Completed work has approved spending, recorded costs and a resulting condition. An unrepairable item moves to a reviewed end-of-life process rather than being disposed of automatically.

UC-11 Decide end-of-life action

Actors: Department representative or Asset custodian proposes and records completion; Sustainability officer reviews; Finance officer approves write-off; AI service optionally provides advice.

Related requirements: FR-14.1 to FR-14.7, AI-06, AI-11 to AI-13; BR-02, BR-08, BR-09, BR-15.

Precondition: An authorised user has identified an asset for an end-of-life review or UC-10 has opened a proposal. Its previous state is recorded for restoration if the proposal is rejected.

Main success scenario:

1. The department representative or custodian opens UI-11 and prepares Proposed (EL-01) with the action, quantity, recipient/destination, reason and evidence. The reviewed asset is Retired (AS-06).
2. The system shows the reuse-first order and collects reasons why earlier options are unsuitable. If the desirable AI advice feature is used, its output is shown as a recommendation for review.
3. The proposal moves to Under sustainability review (EL-02). The Sustainability officer checks the evidence and any special-handling/EHS consultation and data-erasure record required by BR-09.
4. The eligible Sustainability officer approves the review, and the system sends the proposal to Awaiting finance approval (EL-03).
5. The eligible Finance officer approves the write-off. The system sets Approved (EL-04) and enables completion recording.
6. After the approved donation, disassembly, recycling or disposal is carried out, the custodian records date, responsible person, quantity, destination and evidence. The system sets Completed (EL-05), the appropriate final asset state AS-07 to AS-10, and removes that quantity from availability.

Alternative and exception flows:

2a. AI advice is unavailable or rejected: the authorised staff continue the classic evidence-based review; no recommendation is treated as permission.

3a. Required evidence is missing: the reviewer requests changes, returning the proposal to EL-01. Approval remains blocked until the required records are present.

4a. The action is rejected: the system sets Rejected (EL-06), records the reason and restores the previous asset state. If reuse or repair remains suitable, the appropriate transfer or maintenance workflow can then be used.

5a. Finance rejects the write-off: the proposal follows EL-06 and no final action is authorised.

6a. The action is disassembly: usable parts are registered through UC-02 as new assets linked to the original record.

Postcondition: A completed final action has independent Sustainability and Finance decisions and completion evidence. Rejected proposals do not authorise disposal and restore the asset's previous state.

UC-12 Ask the campus assistant

Actors: Staff user; AI service (supporting).

Related requirements: AI-07, AI-08, AI-11 to AI-13; FR-02.2, FR-07.1 to FR-07.3; BR-01, BR-03, BR-15.

Precondition: The user is signed in with an authorised department. The desirable assistant feature is enabled and the AI service is available for the main path.

Main success scenario:

1. The user opens UI-12 and asks a resource, authorised asset-history, request-status or guidance question.
2. The system limits the available context to records and approved guidance the user's role and department may access.
3. The AI service returns an answer. The system labels it as generated, shows supporting records where available and identifies missing information or uncertainty.
4. The user inspects the answer and may ask a follow-up question. The system records the interaction and user response as required by the AI tracking requirements.

Alternative and exception flows:

2a. The question asks for restricted data: the system declines to disclose it and explains the permitted scope without revealing the protected record.

3a. The service is unavailable, times out or returns an unsupported answer: the system preserves the user's input and offers classic search, forms or help.

3b. The assistant proposes a resource request or record change: the system shows the exact proposed fields. The user accepts, corrects or cancels them; only explicit confirmation passes them to the normal validated workflow. A submitted request still becomes Pending department approval under UC-04.

4a. The user asks the assistant to approve a transfer or dispose of an item: the system offers guidance only and leaves the decision to the assigned human role through UC-08 or UC-11.

Postcondition: The user receives a reviewable answer or a classic alternative. An assistant response alone changes no record, role, ownership or approval state.

UC-13 Generate and review report

Actors: Business administrator, Finance officer, Procurement officer, Sustainability officer; AI service supports generated narrative.

Related requirements: FR-17.1 to FR-17.5, AI-09 to AI-13, SU-01 to SU-09; BR-10, BR-15, BR-18.

Precondition: The user is signed in and may view the chosen departments and report data. Source transactions and any indicator inputs are available or their absence can be reported.

Main success scenario:

1. The user opens UI-13 and selects a report type, period and authorised departments.
2. The system produces the conventional report with its filters, record counts, totals, units, source records and limitations, marked Draft.
3. If a narrative summary is requested, the system sends authorised report data to the AI service and displays the generated text as a draft with supporting sources, model information and uncertainty where available.
4. The relevant data owner checks the figures and any narrative against the conventional totals and records corrections. Finance reviews costs/savings, Sustainability reviews sustainability figures, and the Business administrator reviews operational figures.
5. After the required data-owner review, the system marks the report Reviewed by with the reviewing user and date. A report covering several data domains remains Draft until each responsible owner has checked their figures.

Alternative and exception flows:

2a. An indicator input or defensible factor is missing: the system shows N/A or an explicitly identified estimate, not an invented saving or carbon result.

3a. AI is unavailable: the conventional report remains usable and no generated narrative is required to view its totals.

4a. A generated claim is unsupported or its total differs: the data owner rejects or corrects the draft; it cannot become an official summary while the inconsistency remains.

4b. A Procurement officer or other viewer is not the required data owner: they may inspect authorised data and provide comments, but cannot substitute for the BR-10 review.

5a. Desirable export is available: the user may obtain the authorised CSV or printable report with its filters and review status.

Postcondition: A traceable report is available with its actual review status. Official figures derive from the defined records and formulas; unaccepted AI suggestions and unfulfilled requests do not count as realised benefits.

UC-14 View asset history and audit log

Actors: Asset custodian, Business administrator; other staff only where their role permits access to the underlying record.

Related requirements: FR-15.1, FR-15.2, FR-18.1 to FR-18.4, FR-02.2; BR-17.

Precondition: The user is signed in and may view the asset history or, for the Business administrator, the audit records within their assigned scope.

Main success scenario:

1. The custodian opens an authorised asset's history in UI-04, or the Business administrator opens the audit view.
2. The system displays the recorded events with date, actor, action, state changes, approvals, costs and available evidence; the audit view also shows reasons and before/after values.
3. The user filters the history by event type and period. The Business administrator may search audit records by user, record, action and period.
4. The user opens a supporting record. The system checks access again before displaying its details.
5. The Business administrator may export the authorised audit results as CSV under FR-18.3. The system retains the original entries for the TBD-7 period.

Alternative and exception flows:

2a. No events match: the system explains the empty result and keeps the filters available.

4a. A supporting record is outside the user's access: the system refuses disclosure even if a link is visible elsewhere.

5a. Any user tries to edit or delete an audit entry: the system refuses. A correction is a new traceable entry referring to the original under BR-17.

Postcondition: The user has inspected permitted history or audit evidence. Reading or exporting it does not alter the underlying asset, decisions or original audit entries.

4.2 Record States

The project description asks for the states of assets, requests, transfers and maintenance. These tables list every state, what it means and what can come next. A record is always in exactly one state. If someone tries a change that isn't in the "Next states" column, the system blocks it and logs the attempt (FR-05.2). The state names are the labels users see on screen, and we will turn these tables into UML state diagrams in Phase 5.

4.2.1 Asset Life-Cycle States

How many units are available, reserved or allocated is tracked separately from the state (FR-05.3). That way a batch of chairs can be in service while some of them are listed for reuse.

ID	State	Meaning	Next states
AS-01	Draft	Record being entered; not yet confirmed	AS-02, AS-03 (on confirmation)
AS-02	In service	In use by the owning department	AS-03, AS-04, AS-05, AS-06
AS-03	In storage	Not in use; may be listed for reuse	AS-02, AS-04, AS-05, AS-06

ID	State	Meaning	Next states
AS-04	Under maintenance	An approved work order is open	AS-02, AS-03, AS-06 (not repairable)
AS-05	In transfer	Handed over by the releasing custodian, awaiting receipt	AS-02 or AS-03 under the new owner (receipt confirmed); previous state (transfer cancelled or dispute resolved)
AS-06	Retired	Out of service; an end-of-life action is proposed or under review	AS-02, AS-03 (action rejected), AS-07, AS-08, AS-09, AS-10
AS-07	Donated	Given to an approved recipient	None (final)
AS-08	Recycled	Sent to an approved recycler	None (final)
AS-09	Disassembled	Taken apart; usable parts registered as new assets	None (final)
AS-10	Disposed	Disposed of as the last option	None (final)

4.2.2 Listing States

ID	State	Meaning	Next states
LS-01	Draft	Listing being prepared by the custodian or department representative	LS-02, deleted
LS-02	Pending release approval	Waiting for the releasing department representative (BR-04)	LS-03, LS-01 (changes requested), LS-05
LS-03	Published	Visible to authorised staff; quantity can be reserved	LS-04, LS-05, LS-06
LS-04	Fully reserved	All offered quantity is held by reservations	LS-03 (a reservation ends), LS-06
LS-05	Withdrawn	Removed by the owning department (not allowed during an approved transfer, BR-19)	LS-01
LS-06	Closed	All offered quantity has been transferred	None (final)

4.2.3 Request States

ID	State	Meaning	Next states
RQ-01	Draft	Being written by the requester or drafted by the assistant	RQ-02, deleted
RQ-02	Pending department approval	Waiting for the requester's department representative (BR-03)	RQ-03, RQ-01 (changes requested), RQ-08
RQ-03	Open	Approved; the requester searches and reviews matches	RQ-04, RQ-07, RQ-09, RQ-10, RQ-02 (changed after approval)
RQ-04	Reservation requested	A listing has been selected and a reservation requested	RQ-05, RQ-03 (reservation declined or expired), RQ-09
RQ-05	In transfer	A transfer for the request is being approved or carried out	RQ-06, RQ-03 (transfer rejected or cancelled)
RQ-06	Fulfilled	The requested quantity has been received	None (final)

ID	State	Meaning	Next states
RQ-07	No suitable item	Closed because no eligible asset exists; the requester may follow the normal purchasing process outside the system	None (final)
RQ-08	Rejected	Refused by the department representative, with a reason	None (final)
RQ-09	Cancelled	Withdrawn by the requester	None (final)
RQ-10	Expired	The required-by date passed before fulfilment	None (final)

4.2.4 Reservation States

ID	State	Meaning	Next states
RV-01	Requested	The requester asked to hold a quantity	RV-02, RV-03, RV-06
RV-02	Confirmed	The releasing custodian or representative holds the quantity until the hold period ends (BR-12)	RV-04, RV-05, RV-06
RV-03	Declined	The releasing side refused, with a reason	None (final)
RV-04	Converted to transfer	A transfer record was created from the reservation	None (final)
RV-05	Expired	The hold period ended without a transfer	None (final)
RV-06	Cancelled	The requester or the releasing side cancelled it	None (final)

4.2.5 Transfer States

ID	State	Meaning	Next states
TR-01	Pending release approval	Waiting for the releasing department representative (BR-05)	TR-02, TR-08, TR-09
TR-02	Pending receipt approval	Waiting for the receiving department representative (BR-05)	TR-03, TR-04, TR-08, TR-09
TR-03	Pending policy approval	Waiting for the Finance and/or Procurement officer, only when BR-06 applies	TR-04, TR-08, TR-09
TR-04	Approved	All approvals given; collection or delivery is being arranged	TR-05, TR-09
TR-05	Handed over	The releasing custodian confirmed handover, quantity and condition	TR-06, TR-07
TR-06	Completed	The receiving custodian confirmed receipt; ownership, custody and location updated (BR-13)	None (final)
TR-07	Disputed	The receiving custodian recorded a quantity or condition difference	TR-06 (resolved and received), TR-09 (returned)
TR-08	Rejected	An approver refused the transfer, with a reason	None (final)
TR-09	Cancelled	Cancelled before completion; the held quantity returns to the listing	None (final)

4.2.6 Maintenance Work Order States

ID	State	Meaning	Next states
MW-01	Reported	A defect or inspection request was reported	MW-02, MW-09
MW-02	Under assessment	Maintenance staff inspect the asset and estimate the cost	MW-03, MW-07, MW-09
MW-03	Awaiting cost approval	Waiting for approval of the estimate (BR-07)	MW-04, MW-08
MW-04	Approved	Spending approved; work can start	MW-05, MW-09
MW-05	In progress	Repair or refurbishment under way	MW-06, MW-07, MW-03 (cost exceeds the approved amount)
MW-06	Completed	Work, actual cost, outcome and new condition recorded; the asset returns to service or storage	None (final)
MW-07	Not repairable	Repair is impossible or uneconomic; an end-of-life review starts (EL-01)	None (final)
MW-08	Rejected	Spending was not approved, with a reason	None (final)
MW-09	Cancelled	The work order was withdrawn	None (final)

4.2.7 End-of-Life Action States

ID	State	Meaning	Next states
EL-01	Proposed	Action (retain, donate, disassemble, recycle or dispose), reason and evidence entered; an AI recommendation may be attached	EL-02
EL-02	Under sustainability review	The Sustainability officer checks the action against the reuse-first order (BR-08) and any EHS rules (BR-09)	EL-03, EL-06, EL-01 (changes requested)
EL-03	Awaiting finance approval	The Finance officer approves the write-off	EL-04, EL-06
EL-04	Approved	The action may be carried out	EL-05
EL-05	Completed	Date, recipient or destination, quantity and evidence recorded; the asset enters its final state	None (final)
EL-06	Rejected	The asset returns to its previous state	None (final)

4.2.8 Approval Task States

ID	State	Meaning	Next states
AT-01	Pending	Assigned to the users who hold the required role	AT-02, AT-03, AT-04, AT-05
AT-02	Approved	Decision recorded with user, role and time	None (final)
AT-03	Rejected	Decision recorded with a mandatory reason	None (final)
AT-04	Changes requested	Returned to the submitter with a reason	None (final; a new task is created on resubmission)
AT-05	Withdrawn	The parent record was cancelled before a decision	None (final)

4.3 User Accounts and Access Control

4.3.1 Description and Priority

Staff log in and only see what their roles allow. A person can have more than one role, each tied to a department. Giving someone a role takes two people: a Business administrator approves it and a System administrator sets it up, so IT access never turns into approval power. Based on Phase 1 FR-01 and FR-02. Use case UC-01. **Priority: Essential (Must, High).**

4.3.2 Stimulus/Response Sequences

1. *Business administrator* requests a role for a user in a department. *System* records the request as waiting for authorisation.
2. *Business administrator* authorises; *System administrator* applies it. *System* activates the role and writes an audit entry.
3. *User* signs in. *System* shows the dashboard with the user's tasks and notifications for all their roles.
4. *User* attempts an action their roles do not allow. *System* refuses it, names the role required and logs the attempt.

4.3.3 Functional Requirements

ID	Requirement
FR-01.1	The system shall allow the System administrator to create, update and deactivate user accounts with name, staff email, department and account status.
FR-01.2	The system shall allow one user to hold several of the nine roles, each limited to one or more named departments.
FR-01.3	When a role assignment is requested, the system shall require authorisation by a Business administrator before a System administrator can apply it (BR-01).
FR-01.4	When an account is deactivated, the system shall end its active sessions and list its open approval tasks for reassignment by the Business administrator.
FR-02.1	The system shall require authentication before showing any page other than the sign-in page.
FR-02.2	The system shall check the user's role and department on the server for every protected action, including actions proposed through the campus assistant.
FR-02.3	If a user attempts an action their roles do not permit, then the system shall refuse it, state the role required and record the attempt in the audit log.
FR-02.4	If five consecutive sign-in attempts fail for one account, then the system shall block further attempts on that account for 15 minutes.

4.4 Departments, Categories and Approval Settings

4.4.1 Description and Priority

Business administrators keep the basic lists up to date: departments and who represents them, asset categories, and the limits used for approvals and reservations. Based on Phase 1 FR-03. Use case UC-01. **Priority: Essential (Must, High)**, except FR-03.3, which is Desirable. Until it's built, the proposed values in Appendix C are set in the configuration.

4.4.2 Stimulus/Response Sequences

1. *Business administrator* adds a department and names its representatives and custodians. *System* makes the department available for assets, requests and approval routing.
2. *Business administrator* adds or deactivates a category. *System* updates the category list used by forms, search and AI classification.

4.4.3 Functional Requirements

ID	Requirement
FR-03.1	The system shall allow a Business administrator to create and update departments with a name, a code, department representatives and asset custodians.
FR-03.2	The system shall allow a Business administrator to maintain categories and subcategories, each with a counting unit (item, set or kilogram) and an optional emission-factor reference (SU-08).
FR-03.3	The system shall allow a Business administrator to change the approval thresholds (TBD-1, TBD-2), the reservation hold period (TBD-3), the verification period (TBD-4) and the reminder period (TBD-5), recording each change with an effective date; a change shall not alter approvals already in progress. Desirable (Should).
FR-03.4	The system shall allow a Business administrator to mark a category as procurement-controlled (BR-06) or as requiring special handling (BR-09).
FR-03.5	If a Business administrator tries to delete a category or department that existing records use, then the system shall refuse and offer to deactivate it instead.

4.5 Asset Registration and Records

4.5.1 Description and Priority

Custodians and department representatives register assets with all the details the project description asks for, confirm them and keep them up to date. Each asset has a state and, separately, counts of how many units are available, reserved and allocated. The AI suggestions for category and description are in 4.17. Based on Phase 1 FR-04 and FR-05. Use case UC-02. **Priority: Essential (Must, High).**

4.5.2 Stimulus/Response Sequences

1. *Custodian* enters the asset details, photographs and documents. *System* validates the fields and requests AI suggestions (4.17).
2. *Custodian* accepts, edits or rejects each suggestion and confirms the record. *System* saves it as In service or In storage and records who confirmed it.
3. *Custodian* updates the condition or location later. *System* keeps the previous values in the asset history.

4.5.3 Functional Requirements

ID	Requirement
FR-04.1	The system shall allow a custodian or department representative to register an asset with a system-generated unique ID, name, description, category and subcategory, quantity and unit, owning department, custodian, location (building and room), purchase date, purchase or estimated value, condition (New, Good, Fair, Poor, Not working), technical specifications, up to five photographs (JPG or PNG, 5 MB each) and supporting documents (PDF, 10 MB each).

ID	Requirement
FR-04.2	If a mandatory field is missing or invalid, then the system shall keep the asset in Draft (AS-01) and show which fields need correction.
FR-04.3	When the custodian or department representative confirms the record, the system shall save it as In service (AS-02) or In storage (AS-03) and record the confirming user and time.
FR-04.4	The system shall record the date and the user of each asset's most recent availability and condition verification.
FR-04.5	When a new asset has the same department, category, name and specifications as an existing asset, the system shall warn of a possible duplicate and require confirmation before saving.
FR-04.6	When a confirmed asset record is edited, the system shall keep the previous values in the asset's history (FR-15.1).
FR-05.1	The system shall keep each asset in exactly one life-cycle state from section 4.2.1.
FR-05.2	If a user or process attempts a state change that section 4.2 does not allow, then the system shall reject it and record the attempt in the audit log.
FR-05.3	The system shall track the available, reserved and allocated quantity of each asset so that their sum never exceeds the total quantity (BR-11).
FR-05.4	While an asset is Under maintenance, In transfer or Retired, the system shall not allow its quantity to be published or reserved.

4.6 Resource Publication

4.6.1 Description and Priority

A department offers an item it doesn't need to the rest of campus by listing it. The department representative has to approve the listing before anyone else can see it. Based on Phase 1 FR-06. Use case UC-03. **Priority: Essential (Must, High).**

4.6.2 Stimulus/Response Sequences

1. *Custodian* creates a listing for an asset in service or in storage. *System* checks the quantity and sends the listing for release approval.
2. *Department representative* approves. *System* publishes the listing to all authorised staff and notifies the custodian.
3. *Custodian* withdraws the listing. *System* cancels pending reservations and notifies the requesters.

4.6.3 Functional Requirements

ID	Requirement
FR-06.1	The system shall allow a custodian or department representative to create a listing for an asset that is In service or In storage, stating the quantity offered, condition, location, available-from date, restrictions (for example "collection required") and reason (surplus, underutilised or unwanted).
FR-06.2	When a listing is submitted, the system shall create a release-approval task for the owning department's representative and set the listing to Pending release approval (LS-02, BR-04).
FR-06.3	When the release is approved, the system shall publish the listing (LS-03) to authorised staff, showing quantity, condition, location, restrictions, custodian and the last verification date.
FR-06.4	If the asset's last verification is older than the verification period (TBD-4), then the system shall show "Verification needed" on the listing and block reservation confirmation until the custodian re-verifies it (BR-16).

ID	Requirement
FR-06.5	The system shall allow the owning department to withdraw a listing (LS-05) unless an approved transfer for it is in progress (BR-19); withdrawal shall cancel its pending reservations and notify the requesters.
FR-06.6	When the whole offered quantity has been transferred, the system shall close the listing (LS-06).

4.7 Resource Requests

4.7.1 Description and Priority

A requester describes what their department needs. Once the department representative approves it, the requester can search, look at matches and reserve an item. The request always shows its status and who has to act next. Based on Phase 1 FR-07. Use case UC-04. **Priority: Essential (Must, High).**

4.7.2 Stimulus/Response Sequences

1. *Requester* fills in the request form or confirms an assistant draft. *System* validates it and sends it for department approval.
2. *Department representative* approves. *System* opens the request and shows eligible matches (4.18–4.19).
3. *Requester* changes the quantity, specification or date of an open request. *System* returns it for department approval.
4. *Requester* finds no suitable item and closes the request. *System* records the outcome as No suitable item.

4.7.3 Functional Requirements

ID	Requirement
FR-07.1	The system shall allow a requester to create a request with category, purpose, technical specifications, quantity, minimum acceptable condition, urgency (Low, Normal, High), delivery location, required-by date and an optional plain-language description.
FR-07.2	If the required-by date is in the past or the quantity is not a positive whole number, then the system shall refuse submission and explain the error.
FR-07.3	When a request is submitted, the system shall set it to Pending department approval (RQ-02) and create an approval task for the requester's department representative (BR-03).
FR-07.4	If the quantity, specifications or required-by date of an approved request change, then the system shall return the request to Pending department approval (RQ-02).
FR-07.5	The system shall show the requester the current state, the role responsible for the next step and every decision taken on the request.
FR-07.6	The system shall allow the requester to cancel a request before it is Fulfilled; cancellation shall release any reservation linked to it (BR-20).
FR-07.7	When the required-by date passes before fulfilment, the system shall set the request to Expired (RQ-10), release any linked reservation and notify the requester.
FR-07.8	When the requester closes a request because no eligible asset exists, the system shall record it as No suitable item (RQ-07), state that the normal purchasing process may proceed outside Dawra Campus, and exclude it from avoided-purchase figures (BR-18).

4.8 Search and Filtering

4.8.1 Description and Priority

Staff search listed items by keyword and filters. This normal search always works, with or without AI (AI-13). The AI matching is in 4.18. Based on Phase 1 FR-08. Use case UC-05. **Priority: Essential (Must, High).**

4.8.2 Stimulus/Response Sequences

1. *User* enters keywords and selects filters. *System* returns the authorised listings that match, with the number of results.
2. *User* finds nothing. *System* says no listing matches and offers to start a request with the same criteria.

4.8.3 Functional Requirements

ID	Requirement
FR-08.1	The system shall search the name, description, category, tags and specifications of published listings by keyword.
FR-08.2	The system shall filter results by category, condition, department, location, available quantity, asset age, estimated value and life-cycle state, and shall allow filters to be combined.
FR-08.3	The system shall sort results by newest listing, condition or available quantity.
FR-08.4	The system shall return only records the user is authorised to see; unpublished listings shall be visible only to the owning department.
FR-08.5	When no listing matches, the system shall say so and offer to create a request with the same criteria.
FR-08.6	The keyword search and filters shall remain available while the AI services are unavailable.

4.9 Reservations

4.9.1 Description and Priority

A requester asks to hold some units of a listed item. The releasing department confirms the hold, which lasts a set time and then runs out unless a transfer gets approved. Reserving an item is not the same as approving the transfer. Based on Phase 1 FR-09. Use case UC-07. **Priority: Essential (Must, High).**

4.9.2 Stimulus/Response Sequences

1. *Requester* selects a listing and a quantity for an open request. *System* checks availability and records the reservation as Requested.
2. *Custodian or department representative* confirms. *System* holds the quantity and shows the expiry date to both sides.
3. The hold period ends without a transfer. *System* releases the quantity and notifies both sides.

4.9.3 Functional Requirements

ID	Requirement
FR-09.1	The system shall allow a requester with an Open request to request a reservation of a quantity from a Published listing (RV-01).

ID	Requirement
FR-09.2	If the quantity requested exceeds the available quantity, then the system shall refuse the reservation and show the quantity available (BR-11).
FR-09.3	When the releasing custodian or department representative confirms a reservation, the system shall hold the quantity until the end of the hold period (TBD-3) and reduce the available quantity (RV-02).
FR-09.4	When a hold period ends without an approved transfer, the system shall release the quantity (RV-05) and notify the requester and the custodian (BR-12).
FR-09.5	When two users try to reserve the last available units at the same time, the system shall confirm only one reservation and tell the other user that the quantity is no longer available.
FR-09.6	When a confirmed reservation becomes a transfer, the system shall link the two records and set the reservation to Converted to transfer (RV-04).
FR-09.7	The system shall state on every reservation that the item is held, not yet approved for transfer (BR-15).

4.10 Approval Workflow

4.10.1 Description and Priority

Every approval in section 5.5 becomes a task sent to the people with the right role in that department. Approvers see everything they need in one inbox, and every decision is saved with its reason. Based on Phase 1 FR-10. Use case UC-08. **Priority: Essential (Must, High).**

4.10.2 Stimulus/Response Sequences

1. A record reaches an approval stage. *System* creates a task for the required role and notifies the approvers.
2. *Approver* opens the task. *System* shows the record summary, the requester, the asset, costs, any AI score and explanation, and the sustainability effect.
3. *Approver* approves, rejects or requests changes. *System* records the decision and moves the record to its next state.

4.10.3 Functional Requirements

ID	Requirement
FR-10.1	The system shall create an approval task for each stage in Table 5.2 and assign it to the users holding the required role for the relevant department.
FR-10.2	The system shall show each approver one inbox listing pending tasks with the record summary, requester, asset, quantity, costs, any AI score and explanation, and sustainability effect.
FR-10.3	The system shall allow an approver to approve, reject or request changes; if the approver rejects or requests changes, then the system shall require a reason (BR-20).
FR-10.4	If a user is the requester or creator of the record being approved, then the system shall not allow that user to approve it (BR-02).
FR-10.5	When an approval task has been pending for longer than the reminder period (TBD-5), the system shall remind the approvers and mark the task as overdue on their dashboard.
FR-10.6	The system shall record each decision with the approver, role, date and time, reason and the version of the record approved.
FR-10.7	If an approved record is changed in a way that affects the decision (quantity, value, recipient or action), then the system shall withdraw the approval and create a new task.
FR-10.8	The system shall allow a Business administrator to reassign a task to another user who holds the required role, without granting the administrator approval authority (BR-01).

4.11 Transfer, Handover and Receipt

4.11.1 Description and Priority

A transfer moves an item from one department to another. Both departments have to approve it, plus Finance or Procurement when a limit applies. The owner and location only change once the receiving custodian confirms they got the item. Based on Phase 1 FR-11. Use case UC-09. **Priority: Essential (Must, High).**

4.11.2 Stimulus/Response Sequences

1. A reservation is confirmed and the requester proceeds. *System* creates the transfer and starts the approvals in Table 5.2.
2. All approvals are given. *System* asks both custodians to arrange collection or delivery.
3. *Releasing custodian* confirms handover, quantity and condition. *System* marks the asset In transfer.
4. *Receiving custodian* confirms receipt. *System* updates ownership, custody and location, completes the transfer and fulfils the request.
5. *Receiving custodian* reports a difference. *System* marks the transfer Disputed and notifies both department representatives.

4.11.3 Functional Requirements

ID	Requirement
FR-11.1	When a requester proceeds with a confirmed reservation, the system shall create a transfer record with the asset, quantity, releasing and receiving departments and custodians, collection or delivery arrangement, planned date and supporting documents.
FR-11.2	The system shall require release approval and receipt approval (BR-05), and policy approval where BR-06 applies, before setting a transfer to Approved (TR-04).
FR-11.3	When the releasing custodian confirms handover with quantity and condition, the system shall set the transfer to Handed over (TR-05) and the asset to In transfer (AS-05).
FR-11.4	When the receiving custodian confirms receipt of the agreed quantity and condition, the system shall set the transfer to Completed (TR-06), change the owning department, custodian and location, and set the linked request to Fulfilled (BR-13).
FR-11.5	If the receiving custodian records a quantity or condition different from the handover record, then the system shall set the transfer to Disputed (TR-07), keep ownership unchanged and notify both department representatives.
FR-11.6	The system shall allow a transfer to be cancelled before handover and shall then return the held quantity to the listing (TR-09).
FR-11.7	The system shall produce a printable and exportable transfer record with all approvals and confirmations, for reconciliation with the existing KU asset register.

4.12 Inspection, Maintenance and Repair

4.12.1 Description and Priority

Staff report problems and record inspections. Maintenance staff check the item, estimate the cost, do the approved work and record what happened. If an item can't be fixed, it goes to the end-of-life review. Based on Phase 1 FR-12 and FR-13. Use case UC-10. **Priority: Essential (Must, High).**

4.12.2 Stimulus/Response Sequences

1. *Custodian* reports a defect. *System* opens a work order and notifies maintenance staff.
2. *Maintenance staff* record the assessment and estimated cost. *System* sends the estimate for cost approval (BR-07).
3. *Maintenance staff* complete the work. *System* records the cost and outcome, updates the condition and returns the asset to service.
4. *Maintenance staff* mark the asset not repairable. *System* starts an end-of-life review.

4.12.3 Functional Requirements

ID	Requirement
FR-12.1	The system shall allow authorised staff to record an inspection with the asset, date, inspector, findings, condition rating, photographs and recommended follow-up (no action, repair, refurbish or end-of-life review).
FR-12.2	When an inspection rates an asset Not working or unsafe, the system shall block its publication and reservation until it is repaired or an end-of-life decision is made.
FR-13.1	The system shall allow a custodian or department representative to report a defect, which opens a work order in state Reported (MW-01).
FR-13.2	The system shall allow maintenance staff to record the assessment, the estimated cost and the planned work, and shall then route the estimate for approval under BR-07 (MW-03).
FR-13.3	While a work order is Approved or In progress, the system shall keep the asset Under maintenance (AS-04).
FR-13.4	The system shall allow maintenance staff to record the work performed, parts, labour hours, actual cost, outcome and resulting condition.
FR-13.5	If the actual cost exceeds the approved amount, then the system shall require approval of the difference under BR-07 before the work order can be completed.
FR-13.6	When a work order is completed, the system shall update the asset's condition and return it to its previous state (In service or In storage).
FR-13.7	When maintenance staff mark an asset Not repairable, the system shall close the work order (MW-07), set the asset to Retired (AS-06) and open an end-of-life proposal (EL-01).

4.13 Donation, Recycling, Retirement and Disposal

4.13.1 Description and Priority

When an item leaves service, the department suggests what to do with it. The Sustainability officer checks that reuse and repair were considered first, and the Finance officer approves the write-off. Throwing it away is the last option. The AI suggestion in 4.20 can help, but it never decides. Based on Phase 1 FR-14. Use case UC-11. **Priority: Essential (Must, High).**

4.13.2 Stimulus/Response Sequences

1. *Department representative* proposes an action with a reason and evidence. *System* shows any AI recommendation and sends the proposal for sustainability review.
2. *Sustainability officer* approves the review. *System* sends it to the Finance officer.
3. *Finance officer* approves. *System* allows the action and asks for evidence when it is done.
4. *Custodian* records completion with evidence. *System* sets the asset's final state and removes it from availability.

4.13.3 Functional Requirements

ID	Requirement
FR-14.1	The system shall allow a department representative or custodian to propose an end-of-life action (retain, donate, disassemble for parts, recycle or dispose) with the reason, quantity, proposed recipient or destination and evidence (EL-01).
FR-14.2	If the proposed action is recycling or disposal, then the system shall require a recorded reason why reuse, repair and donation are not suitable (BR-08).
FR-14.3	The system shall route each proposal to the Sustainability officer for review and then to the Finance officer for write-off approval (Table 5.2).
FR-14.4	Where the asset's category requires special handling, the system shall require a recorded EHS consultation and, for devices that store data, a data-erasure record before approval (BR-09).
FR-14.5	When an approved action is carried out, the system shall record the date, responsible person, recipient or destination, quantity and evidence, set the asset's final state (AS-07 to AS-10) and remove the quantity from availability.
FR-14.6	If a proposal is rejected, then the system shall return the asset to the state it had before it was retired (EL-06).
FR-14.7	When an asset is disassembled, the system shall allow its usable parts to be registered as new assets linked to the original record.

4.14 Life-Cycle History and Audit Trail

4.14.1 Description and Priority

Every asset has a full history you can read, and every important action goes into an audit log that nobody can change. Based on Phase 1 FR-15 and FR-18. Use case UC-14. **Priority: Essential (Must, High).**

4.14.2 Stimulus/Response Sequences

1. *Custodian* opens an asset. *System* shows its history from registration to the present, newest first.
2. *Business administrator* searches the audit log by user and period. *System* lists the matching entries and allows export.

4.14.3 Functional Requirements

ID	Requirement
FR-15.1	The system shall show each asset's chronological history: registration, edits, listings, reservations, transfers, inspections, maintenance and end-of-life actions, each with date, actor, approvals, costs and evidence.
FR-15.2	The system shall allow the history to be filtered by event type and period.

ID	Requirement
FR-18.1	The system shall write an audit entry for every creation, change, state change, approval decision, role change, refused action and response to an AI output, recording the actor, role, time, record, action, reason and the values before and after.
FR-18.2	The system shall not allow any role to edit or delete an audit entry; corrections shall be recorded as new entries (BR-17).
FR-18.3	The system shall allow a Business administrator to search the audit log by user, record, action and period, and to export the results as CSV.
FR-18.4	The system shall keep audit entries for the retention period (TBD-7).

4.15 Notifications

4.15.1 Description and Priority

Users get a notification in the app whenever something needs their action or something they're involved in changes. Based on Phase 1 FR-16. **Priority: Essential (Must, High)**; FR-16.6 is a Future enhancement.

4.15.2 Stimulus/Response Sequences

1. A task is assigned to a user. *System* shows a notification on the user's dashboard with a link to the task.
2. *User* opens the notification. *System* opens the record and marks the notification as read.

4.15.3 Functional Requirements

ID	Requirement
FR-16.1	When an approval task is created or reassigned, the system shall notify the users who can decide it.
FR-16.2	When a request, reservation, transfer, work order or end-of-life action changes state, the system shall notify the requester and the custodians and representatives involved.
FR-16.3	When a reservation is one working day from expiry, the system shall notify the requester and the custodian.
FR-16.4	The system shall show the number of unread notifications on every page, link each notification to its record and allow it to be marked as read.
FR-16.5	The system shall show a notification only to users who are allowed to view the record it refers to.
FR-16.6	Where email delivery is enabled, the system shall also send each notification by email. Future enhancement (Could).

4.16 Conventional Reports and Dashboards

4.16.1 Description and Priority

The normal reports give exact numbers that come out the same every time. The AI summaries in 4.22 and the sustainability numbers in 6.1 use these figures. A report only becomes official after the person responsible for that data checks it. Based on Phase 1 FR-17. Use case UC-13. **Priority: Essential (Must, High)**; FR-17.3 is Desirable.

4.16.2 Stimulus/Response Sequences

1. *Finance officer* selects a report, a period and departments. *System* produces it with totals and the filters used, marked Draft.
2. *Finance officer* reviews it. *System* marks it "Reviewed by" with the name and date.

4.16.3 Functional Requirements

ID	Requirement
FR-17.1	The system shall produce reports on: assets by department and state; available listings; open requests and their age; reservations; approvals (pending, overdue, decided); completed transfers and ownership changes; maintenance history and costs; end-of-life actions; and the sustainability indicators in section 6.1.
FR-17.2	The system shall filter every report by period and department and show the filters, record counts and totals used.
FR-17.3	The system shall export reports as CSV and provide a printable view. Desirable (Should).
FR-17.4	The system shall mark each report Draft until its data owner reviews it, then show the reviewer's name and date (BR-10).
FR-17.5	The system shall show each user a dashboard with their pending tasks, their requests, reservations close to expiry and assets needing verification.

4.17 AI Asset Classification

4.17.1 Description and Priority

AI helps the Asset custodian or Department representative complete an asset record by suggesting categories and other details. The user checks each suggestion before it becomes part of the record. This feature refines Phase 1 AI-01.

Priority: Essential (Must, High).

4.17.2 Stimulus/Response Sequences

1. The Asset custodian or Department representative enters asset details and asks for suggestions. The system checks the supplied information and requests AI assistance.
2. The system displays the suggested details, reasons and confidence information, with any missing or uncertain information clearly marked.
3. The user accepts, corrects or rejects the suggestions. The system applies only the confirmed details and records the user's decision.
4. If AI is unavailable or cannot read an attachment, the system explains the issue and allows manual entry.

4.17.3 Functional Requirements

ID	Requirement
AI-01.1	When a user requests classification assistance, the system shall suggest a category, subcategory, standard description, tags, material where supported, and missing details using the supplied title, description, specifications and supported image or document content.
AI-01.2	If an input format is unsupported or extracted information is uncertain, then the system shall identify the affected input and request manual review rather than treat it as confirmed information.

ID	Requirement
AI-01.3	The system shall display suggested values separately from confirmed asset details and allow the Asset custodian or Department representative to accept, correct or reject them.
AI-01.4	The system shall apply AI suggestions to the asset record only after the authorised user confirms them, while following the asset-record rules in section 4.5 and Table 5.2.
AI-01.5	In a fixed test set of 20 labelled assets, the system shall suggest the correct category for at least 16 assets. Missing, invalid or failed AI responses shall count as incorrect results.

4.18 Semantic Matching and Eligibility

4.18.1 Description and Priority

Semantic matching compares the meaning of a request with asset descriptions, so users can find suitable items even when different words are used. Before AI ranks any results, classic rules check whether each item is eligible. This feature refines Phase 1 AI-02 and AI-03.

Priority: Essential (Must, High).

4.18.2 Stimulus/Response Sequences

1. The Requester asks for matches using a request or search description. The system identifies published listings that the user may access.
2. The system checks the mandatory requirements, removes ineligible items and compares the meaning of the request with the remaining listings.
3. The system presents the eligible matches for ranking and review under section 4.19.
4. If no eligible item exists, the system explains that no suitable match was found and leaves the next action to the user.

4.18.3 Functional Requirements

ID	Requirement
AI-02.1	When a user requests matches, the system shall compare the meaning of the request or search description with permitted published asset listings, including cases where different terms describe the same need.
AI-02.2	For the fixed set of 20 requests with known suitable assets, the system shall return at least one reviewer-labelled useful asset among the first three results for at least 16 requests.
AI-02.3	Using the same 20 requests, asset records and eligibility rules, AI matching shall succeed on at least two more requests than classic keyword search, with success measured by a useful asset appearing in the first three results.
AI-02.4	If no eligible asset exists, then the system shall show a no-match message without presenting an ineligible item as suitable or automatically closing the request. The user may follow FR-07.8 to close it as No suitable item.
AI-03.1	Before AI ranking, the system shall use classic rules to check access rights, availability, quantity, required condition, technical compatibility and the required date, as specified by BR-14.
AI-03.2	When assessing matches, the system shall consider functional purpose, category, technical compatibility, available quantity, condition, location, urgency, required date, estimated transfer or repair cost, and sustainability benefit.

ID	Requirement
AI-03.3	If information needed for a mandatory eligibility check is missing, then the system shall identify the missing information and withhold that item as an actionable recommendation until the check can be completed. Missing costs or sustainability figures shall be marked as unknown rather than assumed to be zero.
AI-03.4	Across the 20-request matching test, the system shall show zero items that fail mandatory eligibility checks as actionable recommendations.

The matching tests use the Phase 1 dataset of 500 synthetic assets, 10 departments and 50 requests. Two of us will label suitable assets independently and resolve differences before fixing the 20-request evaluation set. Classic search and AI matching shall use the same requests and asset snapshot. Failed AI responses count as unsuccessful requests. Cases with no suitable asset shall be tested separately and excluded from the 20-request usefulness calculation. These numbers are acceptance targets, not measured results.

4.19 Ranking, Explanations and Confidence

4.19.1 Description and Priority

The system puts eligible matches in a useful order and explains why each item was recommended. A compatibility score shows how well an item fits the request. Confidence describes the uncertainty of the AI output; it does not give approval or guarantee correctness. This feature refines Phase 1 AI-04 and AI-05.

Priority: Essential (Must, High).

4.19.2 Stimulus/Response Sequences

1. The system receives the eligible matches from section 4.18 and ranks them using documented factors and weights.
2. The Requester opens a recommendation. The system shows its compatibility score, explanation, relevant asset details and confidence information.
3. The Requester chooses the top result or another eligible item. The system records the selection and, when the user overrides the recommended order, the reason.
4. If the user requests a reservation, the system starts the normal reservation workflow under section 4.9.

4.19.3 Functional Requirements

ID	Requirement
AI-04.1	The system shall show each recommended match with a compatibility score from 0 to 100, an explanation of the main matching factors and references to the relevant asset and request details.
AI-04.2	The system shall identify assumptions, missing information and unresolved non-mandatory concerns in the explanation. Mandatory constraints shall remain subject to BR-14.
AI-04.3	The system shall display confidence separately from the compatibility score, explain the basis of the confidence information and mark it as unavailable when no supported measure exists. Neither value shall be presented as a guaranteed probability of correctness.
AI-05.1	The system shall rank eligible matches using documented factors and weights, including how missing information and tied scores are handled.

ID	Requirement
AI-05.2	The system shall allow the user to choose another eligible item instead of the highest-ranked result and shall record the selected item and the user's reason for overriding the recommendation.
AI-05.3	In a review of 20 sampled recommendations, the system shall show the matching reasons, relevant inputs and confidence information or its unavailability for every recommendation.

4.20 Sustainable-Action Recommendation

4.20.1 Description and Priority

AI suggests ways to extend an asset's useful life or handle it responsibly when it leaves service. The recommendation supports the user's decision and follows the reuse-first order in BR-08. This feature refines Phase 1 AI-06.

Priority: Desirable (Should, Medium).

4.20.2 Stimulus/Response Sequences

1. An authorised user requests advice for an asset. The system gathers its permitted condition, demand, repair and cost information.
2. The system suggests an action and explains which information supports it and what remains uncertain.
3. The user accepts the suggestion as a proposal, changes it or rejects it. The system records the decision and keeps the normal approval requirements.
4. If the action concerns an item requiring special handling, the system identifies the checks required by BR-09.

4.20.3 Functional Requirements

ID	Requirement
AI-06.1	When sustainable-action advice is requested, the system shall consider retaining or reusing, transferring, repairing, refurbishing, donating, disassembling, recycling, retiring or disposing of the asset, as relevant to its condition and use.
AI-06.2	The system shall explain how condition, demand, repairability, cost, material, value and sustainability factors affected the recommendation, and identify any missing information.
AI-06.3	The system shall follow the reuse-first order in BR-08 and explain why earlier options are unsuitable before recommending a later option. It shall not invent evidence to justify the recommendation.
AI-06.4	When the user accepts a recommendation, the system shall treat it as a proposal and apply the relevant transfer, maintenance or end-of-life workflow. End-of-life actions shall require Sustainability officer review and Finance officer approval under BR-08.
AI-06.5	Where BR-09 applies, the system shall identify the required EHS consultation and data-erasure evidence. An AI recommendation shall not replace either requirement.
AI-06.6	If the available information is insufficient to support an action, then the system shall state the limitation and request manual review without changing the asset's state.

4.21 Campus Assistant

4.21.1 Description and Priority

The campus assistant lets authorised users ask questions and draft requests in everyday language. It can help users search, understand record status and find relevant guidance. Users must review and confirm any proposed change, and the normal approval rules still apply. This feature refines Phase 1 AI-07 and AI-08.

Priority: Desirable (Should, Medium).

4.21.2 Stimulus/Response Sequences

1. The user asks a question or describes a resource need. The system checks the user's permissions before retrieving relevant information.
2. The assistant answers using permitted records or prepares a draft request with the proposed fields.
3. The user reviews the draft and corrects, confirms or cancels it.
4. After confirmation, the system checks permissions and required fields again. A valid submitted request enters Pending department approval (RQ-02).
5. If information is missing, the action is unauthorised or the user cancels, the system does not submit the request.

4.21.3 Functional Requirements

ID	Requirement
AI-07.1	The system shall support natural-language resource search, asset-history and request-status questions, policy, service and sustainability guidance, and drafting resource requests.
AI-07.2	The system shall limit information used and shown by the assistant to records and guidance the signed-in user is authorised to access.
AI-07.3	When answering a factual question, the system shall provide references to the supporting records or approved guidance and identify any policy values that remain unconfirmed.
AI-07.4	If the available sources do not support an answer, then the system shall state that it cannot confirm the answer rather than invent information.
AI-08.1	Before submitting a request or changing a record through the assistant, the system shall show the exact proposed fields and changes and require explicit user confirmation.
AI-08.2	After confirmation, the system shall recheck permissions and input validity and apply the same approval workflow as the classic interface. A submitted resource request shall enter RQ-02 under BR-03.
AI-08.3	The system shall allow the user to correct or cancel the proposed action before submission. Cancellation shall leave the target record unchanged.
AI-08.4	The assistant shall pass all five planned request-submission tests: confirmed, corrected, cancelled, unauthorised and incomplete requests. Valid submissions shall require confirmation; cancelled, unauthorised and incomplete requests shall not be submitted.

4.22 Generative Reporting

4.22.1 Description and Priority

AI turns conventional report data into a readable draft summary. The conventional reports remain the source of the figures, and the responsible data owners review generated summaries before official use. This feature refines Phase 1 AI-09 and AI-10.

Priority: Essential (Must, High).

4.22.2 Stimulus/Response Sequences

1. An authorised user selects the reporting period, departments and report information. The system produces the conventional figures under section 4.16.
2. The user requests a written summary. The system generates a draft using those figures and permitted supporting records.
3. The user checks the summary and follows its references to the source information.
4. The relevant data owner reviews the summary. The system records the review or keeps the summary in Draft with comments.

4.22.3 Functional Requirements

ID	Requirement
AI-09.1	The system shall generate draft summaries covering resource exchange, high-demand categories, maintenance trends, recommended actions, avoided purchases, financial savings, waste diversion and estimated carbon reductions where supporting data is available.
AI-09.2	The system shall take reported figures from the corresponding conventional reports, using the same record snapshot, filters, units and rounding. Generated totals shall equal the conventional report totals.
AI-09.3	The system shall label generated summaries as AI-generated drafts and distinguish recorded outcomes from estimates, assumptions and proposed actions.
AI-10.1	The system shall link factual claims to supporting records or report totals and identify missing information rather than invent figures or conclusions.
AI-10.2	Before a summary becomes official, the system shall require review by the relevant data owners under BR-10: the Finance officer for financial figures, the Sustainability officer for sustainability figures and the Business administrator for operational figures.
AI-10.3	In five generated test summaries, every factual claim shall be traceable to its supporting information and every reported total shall match the corresponding conventional report. Unsupported claims or mismatched figures shall fail the relevant test.

4.23 Model Tracking, Overrides and Fallback

4.23.1 Description and Priority

The system records how AI outputs were produced and how users responded to them. People retain control over approvals and record changes. If AI fails or is unavailable, users can continue through the classic features. This feature refines Phase 1 AI-11, AI-12 and AI-13.

Priority: Essential (Must, High).

4.23.2 Stimulus/Response Sequences

1. The system receives an AI output and records its source, time and available model information.

2. The user accepts, corrects, rejects or overrides the output. The system records the response and applies the normal validation and approval rules.
3. If an AI service fails, the system shows a clear message, preserves the user's input and makes the relevant classic alternative available.
4. The user continues with classic search, forms, approvals or reports.

4.23.3 Functional Requirements

ID	Requirement
AI-11.1	The system shall treat AI outputs as recommendations and preserve human control over transfers, donation, recycling, disposal, repair spending, financial values and user roles, following the permissions and approval stages in sections 4.2 and 5.5.
AI-11.2	The system shall not treat selecting a match as a reservation, a reservation as transfer approval, or confirmation of an assistant action as approval by another role. A transfer shall reach Approved (TR-04) only after the approvals required by BR-05 and BR-06.
AI-11.3	In the 14 planned approval tests, covering one authorised and one unauthorised path for each of the seven actions in AI-11.1, the system shall allow zero bypasses of required confirmation, permissions or approvals.
AI-12.1	For each AI output, the system shall record the task, input record references, output, generation time, model or deployment identifier, available model version, and any score or confidence information used.
AI-12.2	If model-version or confidence information is unavailable, then the system shall mark it as unavailable rather than create a value.
AI-12.3	When a user accepts, corrects, rejects or overrides an AI output, the system shall record the user, time, original output, response and any corrected values. Corrections, rejections and overrides shall include a reason and follow the audit rules in FR-18 and BR-17.
AI-13.1	If AI is unavailable, rate-limited, returns invalid output or falls below the documented confidence threshold for its task, then the system shall explain the issue and offer the relevant classic search, filters, form or report.
AI-13.2	If an AI request fails or remains unanswered, then the system shall display a failure or timeout message and make the classic alternative available within 15 seconds of the original request, including any retries. It shall preserve entered data and allow the classic workflow to continue.
AI-13.3	The system shall pass all five planned fallback tests: unavailable service, timeout, rate limit, malformed output and insufficient confidence. Each test shall preserve user input and provide the relevant classic alternative within 15 seconds.
AI-13.4	When AI credentials are absent or AI services are disabled, the system shall allow the classic features to start and operate without a paid AI account or AI API key, as required by NFR-09.

5. Other Nonfunctional Requirements

These requirements describe how well the system must work, including its speed, safety, security and ease of use. They retain the Phase 1 requirement IDs. The numerical values are acceptance targets, not results already achieved.

Unless a requirement states otherwise, testing shall use 500 synthetic assets across 10 departments, 50 requests and five concurrent users. We will record the hardware, operating system, browser, software versions and network conditions used so the tests can be repeated.

5.1 Performance Requirements

ID	Requirement	Priority
NFR-02	The system shall display classic search and filter results within 2 seconds for at least 95 of 100 measured searches under the stated test conditions. Time shall be measured from submitting the search until its results are displayed. AI processing time shall be measured separately.	Essential (Must)
NFR-11	With 2,000 assets, 200 requests and five concurrent users, the system shall display classic search results within 3 seconds for at least 95 of 100 measured searches while preserving data integrity. This test shall be reported separately from NFR-02.	Desirable (Should)

For each tested AI request, we will record the response time and whether the request succeeded, failed or timed out. The system shall provide the relevant classic alternative within the 15-second limit in NFR-07 and AI-13.2, including any retries.

Normal search performance and the first request after hosting inactivity shall be measured and reported separately, as agreed in Phase 2. Hosting startup time shall not be presented as AI processing time. The AI fallback test shall begin when the user submits an AI request through the running application. Any proposed change to the acceptance targets requires instructor agreement.

5.2 Safety Requirements

Incorrect records or recommendations could lead to unsafe equipment being transferred or an item being disposed of unnecessarily. The system shall apply the following safeguards:

- If an inspection record identifies an unresolved safety concern, then the system shall block handover until an authorised inspection records that the concern has been addressed for the proposed handling or use. Inspection evidence shall be kept with the asset's maintenance history under section 4.12.
- Before approving donation, recycling or disposal of an asset covered by BR-09, the system shall require the recorded EHS consultation and, for devices that store data, the required data-erasure record.
- When an end-of-life proposal selects a later option in the reuse-first order, the system shall require a recorded reason why the earlier options are unsuitable, as required by BR-08.
- The system shall keep the required human reviews and approvals in place even when an AI recommendation supports the action, following BR-15 and Table 5.2.
- In the agreed safety tests, the system shall block every attempt to bypass a required inspection, special-handling record, reuse-first justification or approval.

5.3 Security Requirements

ID	Requirement	Priority
NFR-03	The system shall check authentication, role and department permissions on the server for every protected action, including assistant actions. Locally managed passwords shall be stored only as salted hashes. Credentials shall remain outside version control, and browser traffic shall use HTTPS as specified in section 3.4. Every unauthorised action in the agreed role-and-permission test matrix shall be blocked.	Essential (Must)

ID	Requirement	Priority
NFR-04	The system shall use synthetic records and test users for the prototype, following C-03 and DR-02. It shall block access to another department's restricted records in every agreed privacy test. AI input shall contain only the approved fields needed for the task. Credentials and unnecessary personal data shall be excluded from AI input content and ordinary logs.	Essential (Must)
NFR-05	The system shall record an audit entry for every action listed in FR-18.1 that occurs in the acceptance tests, including controlled changes, approval decisions, role changes, refused actions and responses to AI outputs. Entries shall identify the actor, role, time, affected record, action, reason and relevant before-and-after values. The system shall reject every tested attempt to edit or delete an audit entry; corrections shall be new entries linked to the original, as required by BR-17.	Essential (Must)

The role-and-permission test matrix shall cover all nine roles, allowed and refused actions, and access within and across departments. It shall also check that technical administration does not grant business approval rights and that users cannot approve their own work, as required by BR-01 and BR-02.

5.4 Software Quality Attributes

ID	Attribute	Requirement	Priority
NFR-01	Usability	The system shall allow at least 4 of 5 representative test users to complete each of three tasks without help: register an asset, find a suitable listing and submit a request. Test users shall have the roles needed for their assigned tasks. Mean post-task satisfaction shall be at least 4 out of 5.	Essential (Must)
NFR-06	Explainability	For all 20 sampled AI recommendations, the system shall show the main reasons, relevant inputs, supporting records and confidence information or its unavailability. It shall distinguish estimates and assumptions from recorded facts, following sections 4.19 and 4.23.	Essential (Must)
NFR-07	Reliability and fallback	In all five planned AI failure tests, the system shall explain the failure and make the relevant classic alternative available within 15 seconds of the original request. The tests shall cover unavailable service, timeout, rate limit, malformed output and insufficient confidence. Every case shall preserve user input and allow the classic workflow to continue with its normal access and approval checks.	Essential (Must)
NFR-08	Maintainability	The system shall separate asset management, workflows, reporting and AI integration into identifiable modules. Project documentation shall explain their responsibilities and interfaces and provide a repeatable command for testing critical business rules.	Essential (Must)
NFR-09	Portability	The system shall run from a fresh repository copy using the documented prerequisites, setup commands, configuration examples and synthetic seed data. One teammate shall complete a fresh setup using those instructions and complete classic search and request workflows without a paid AI account or AI API key.	Essential (Must)

ID	Attribute	Requirement	Priority
NFR-10	Data integrity	The system shall validate quantities, record references and allowed state changes before saving them. Related stock changes shall succeed together or leave the records unchanged. The agreed tests shall produce zero negative quantities, duplicate allocations or invalid state changes. In 10 tests where two users simultaneously attempt to hold the same final unit, exactly one hold shall be confirmed each time.	Essential (Must)
NFR-12	Fairness and ethics	The system shall exclude protected personal characteristics from matching factors and document any department-based policy priorities. In all 10 paired tests that change only personal names or department labels, eligibility and relative ranking shall remain unchanged when permissions, policy priority, asset needs and other relevant factors stay equal.	Essential (Must)
NFR-13	Deployment	The system shall support deployment to the selected hosting environment through documented setup, configuration and restart steps, with a documented HTTPS address. One teammate shall complete a fresh deployment and pass the hosted registration, search, request, approval and AI-fallback checks. Restart testing shall confirm that records, uploaded files and access controls are preserved.	Desirable (Should)

For each requirement, the test record shall identify the requirement ID, test conditions, expected result, actual result and supporting evidence. Unmet targets shall be reported clearly rather than described as completed.

5.5 Business Rules

Business rules are the policies the system has to follow, like who can approve what. They aren't features on their own, but the requirements in section 4 make sure they're followed. Where a rule needs a number from KU, such as a money limit, we use a proposed value until the instructors confirm it (Appendix C).

5.5.1 Rules

ID	Rule
BR-01	Technical administration does not give business approval rights. A Business administrator authorises role assignments and a System administrator applies them; neither approves releases, transfers, spending or disposal unless the user also holds the business role for that department.
BR-02	Nobody approves their own work. The requester or creator of a record cannot decide an approval task for that record. Where a department has only one representative, an alternate approver is named by the Business administrator (TBD-9).
BR-03	A request becomes Open only after the requester's department representative approves it, because receiving an asset makes the department its owner and custodian.
BR-04	A listing becomes visible to other departments only after the owning department's representative approves its release.
BR-05	Every transfer needs two approvals: release by the releasing department's representative and receipt by the receiving department's representative.
BR-06	A transfer also needs Finance officer approval when the asset's value is at or above the policy threshold (TBD-1), and Procurement officer approval when its category is procurement-controlled.
BR-07	Maintenance and repair spending needs the owning department representative's approval, and Finance officer approval when the estimate is at or above the spending threshold (TBD-2).

ID	Rule
BR-08	End-of-life actions follow the reuse-first order: reuse or transfer, repair or refurbish, donate, disassemble for parts, recycle, and only then dispose. Choosing a later option requires a recorded reason why the earlier options are unsuitable. Every action needs Sustainability officer review and Finance officer write-off approval.
BR-09	Assets in special-handling categories (for example electronics with batteries, laboratory equipment with contamination risk, and devices that store data) need a recorded EHS consultation, and data-storing devices need a data-erasure record, before donation, recycling or disposal (TBD-8).
BR-10	A report or generated summary becomes official only after its data owner reviews it: the Finance officer for cost and savings figures, the Sustainability officer for sustainability figures and the Business administrator for operational figures.
BR-11	Available, reserved and allocated quantities may never add up to more than the asset's total quantity, and the same unit may never be reserved or allocated twice.
BR-12	A confirmed reservation is held for the hold period (TBD-3, proposed 5 working days). It expires unless a transfer has been approved by then.
BR-13	Ownership, custody and location change only when the receiving custodian confirms receipt. Until then, the releasing department remains responsible for the asset.
BR-14	Mandatory constraints (access rights, availability, quantity, required condition, technical compatibility and required date) are checked by classic rules before any AI ranking. An asset that fails a mandatory constraint is never shown as an actionable recommendation.
BR-15	AI outputs are recommendations only. Selecting a match is not a reservation, a reservation is not an approval, and an action proposed by the assistant takes effect only after the user confirms it and every normal approval is given. When AI is unavailable, the classic functions are used.
BR-16	An asset whose availability and condition were last verified more than the verification period ago (TBD-4, proposed 90 days) cannot have a reservation confirmed until the custodian verifies it again.
BR-17	Audit entries are never changed or deleted. A mistake is corrected by a new entry that refers to the original.
BR-18	A purchase counts as avoided only when an approved request is fulfilled with an existing asset and an authorised reviewer records that a new purchase would otherwise have been needed. Requests closed as No suitable item never count.
BR-19	A listing cannot be withdrawn while an approved transfer for it is in progress.
BR-20	Every rejection, request for changes and cancellation records a reason, and cancelling or rejecting a record releases any quantity it held.

5.5.2 Approval Stages

Table 5.2 lists every approval in the system, in order. The approver must have the right role for that department, and BR-01 and BR-02 always apply. The states of an approval are in section 4.2.8.

Table 5.2 Approval stages

Workflow	Stages in order and approver	Result when all approve	Result of a rejection
Role assignment	1. Authorise: Business administrator. 2. Apply: System administrator	Role active	Request closed with reason
Asset record	1. Confirm: Asset custodian or department representative	AS-02 or AS-03	Stays in Draft (AS-01)
Listing release	1. Release: owning department representative	Published (LS-03)	Back to Draft (LS-01)

Workflow	Stages in order and approver	Result when all approve	Result of a rejection
Resource request	1. Department approval: requester's department representative	Open (RQ-03)	Rejected (RQ-08)
Reservation	1. Confirm hold: releasing custodian or department representative	Confirmed (RV-02)	Declined (RV-03)
Transfer	1. Release: releasing department representative. 2. Receipt: receiving department representative. 3. Policy, only under BR-06: Finance officer and/or Procurement officer. Then handover and receipt confirmations by the two custodians	Approved (TR-04), then Completed (TR-06)	Rejected (TR-08); held quantity released
Maintenance spending	1. Department: owning department representative. 2. Finance, only at or above TBD-2: Finance officer	Approved (MW-04)	Rejected (MW-08)
End-of-life action	1. Sustainability review: Sustainability officer, including the recorded EHS consultation under BR-09. 2. Write-off: Finance officer	Approved (EL-04), then Completed (EL-05)	Rejected (EL-06); asset returns to its previous state
Official report	1. Review: the data owner under BR-10	Marked Reviewed	Stays Draft with comments

6. Other Requirements

6.1 Sustainability Measurement Requirements

The project description asks us to measure waste diversion, avoided purchases, asset-life extension, reuse, repair, money saved and carbon saved. Table 6.1 shows how each number is calculated. Only finished and approved actions count, never AI suggestions nobody accepted. Since our data is made up, the numbers show how the method works, not real KU savings. Based on Phase 1 SU-01 to SU-08. **Priority: Essential (Must, High)** for SU-01 to SU-09.

Table 6.1 Indicator definitions

Indicator	Calculation for a reporting period	Unit	Source events
Waste diverted	Sum of quantities that left service through reuse, repair returned to service, donation, disassembly or recycling instead of disposal; each item counted once per period; shown by route	Items (kg only where mass is recorded)	TR-06, MW-06, EL-05
Avoided purchases	Number of fulfilled requests confirmed under BR-18, and their value = quantity × replacement cost	Count, AED	TR-06 + reviewer confirmation
Net financial savings	Avoided replacement value − attributable transfer, inspection, repair and refurbishment costs	AED	Avoided purchases, MW-06, TR-06

Indicator	Calculation for a reporting period	Unit	Source events
Reuse rate	Quantity from a defined cohort of listed assets transferred and received by period end ÷ total quantity in that cohort × 100; N/A when the cohort is empty	%	LS-03, TR-06
Repair success rate	Work orders Completed with the asset returned to service ÷ work orders closed as Completed or Not repairable × 100; N/A when none closed	%	MW-06, MW-07
Asset-life extension	Revised retirement date − planned retirement date after reuse, repair or refurbishment, per asset; reported as total and average	Months (estimated)	MW-06, TR-06
Estimated carbon reduction	Σ (avoided quantity × embodied emission factor of the category) − emissions attributed to transport and replacement parts; negative results kept; N/A when no defensible factor exists	kg CO2e (estimated)	Avoided purchases, category factors

ID	Requirement
SU-01	The system shall display waste diverted, avoided purchases, reuse rate, repair success rate, asset-life extension, net financial savings and estimated carbon reduction on a sustainability dashboard and in the conventional reports, filtered by period and department.
SU-02	The system shall calculate waste diverted as defined in Table 6.1, show the result by route (reuse, repair, donation, disassembly, recycling) and count each item or batch quantity once per period.
SU-03	The system shall count an avoided purchase only under BR-18, recording the reviewer, the quantity and the replacement-cost estimate with its source.
SU-04	The system shall calculate net financial savings as defined in Table 6.1, keeping every input value and never counting the same cost or benefit twice.
SU-05	The system shall calculate the reuse rate for a cohort chosen by the user and show the cohort definition, numerator and denominator.
SU-06	The system shall record a planned retirement date for each asset and, after reuse, repair or refurbishment, an estimated revised date, and shall label the resulting life extension as an estimate.
SU-07	The system shall estimate carbon reduction as defined in Table 6.1 using the emission factor recorded for each category, and shall show N/A rather than a figure when a required factor is missing (TBD-6).
SU-08	The system shall show, for each indicator, its formula, source records, factor source and version, assumptions, the reviewing user and whether values are measured, estimated or synthetic.
SU-09	The system shall calculate the repair success rate as defined in Table 6.1 from closed work orders.

6.2 Data Requirements

These are the main things the system stores. We will draw the full entity-relationship diagram in Phase 5.

Record	Main data
User and role assignment	Name, staff email, department, roles per department, account status, authorisation and application of each role
Department and category	Name, code, representatives, custodians; category, subcategory, unit, flags (procurement-controlled, special handling), emission factor reference
Asset	Fields in FR-04.1, life-cycle state, available/reserved/allocated quantities, verification date, planned retirement date

Record	Main data
Listing, request, reservation	Fields in FR-06.1, FR-07.1 and FR-09.1, state, links to each other
Transfer	Fields in FR-11.1, approvals, handover and receipt confirmations, dispute notes
Inspection and work order	Fields in FR-12.1 and FR-13.4, approvals, costs, outcome
End-of-life action	Fields in FR-14.1 and FR-14.5, evidence, EHS consultation, data-erasure record
Approval task, notification, audit entry	Fields in FR-10.6, FR-16 and FR-18.1
AI output record	Task type, input reference, output, score, confidence, model and version, time, user response (Ghaith, section 4.23)
Sustainability factor	Category, factor value, unit, source, version, date

ID	Requirement
DR-01	The system shall include a synthetic data set of at least 500 assets in 10 departments, 50 requests and 20 labelled matching cases, which can be loaded with one command for demonstration and testing.
DR-02	The system shall store no real KU staff personal data or real asset records in this release (C-03).
DR-03	The system shall keep records and audit entries for the retention period (TBD-7) and shall allow a full backup and restore of the database and uploaded files.

6.3 Policy, Legal and Ethical Requirements

These come from the checks we listed in Phase 2 (section 15). They are things we still need to confirm, not a claim that KU has approved the system.

- PL-01: Approval stages, thresholds and special-handling rules shall follow KU asset, procurement and environmental policy once confirmed; until then the proposed values in Appendix C apply and are shown as proposed.
- PL-02: Use of any AI service shall comply with the provider's terms and with KU data rules; no data beyond C-06 shall be sent.
- PL-03: All third-party software shall have a licence that allows academic use and publication of the source code.
- PL-04: The interface and documentation shall state that AI outputs are recommendations and that authorised staff remain accountable for decisions.

6.4 Prioritisation Summary

The project description asks us to group requirements into essential, desirable and future. Everything the description lists as the Phase 6 minimum is essential.

Tier	Features and requirements
Essential prototype functions (Must)	Accounts, roles and access control (4.3); departments and categories (4.4, except FR-03.3); asset registration and states (4.5); publication (4.6); requests (4.7); search and filtering (4.8); reservations (4.9); approval workflow (4.10); transfers (4.11); inspection and maintenance (4.12); end-of-life actions (4.13); history and audit (4.14); notifications (4.15, except FR-16.6); conventional reports and dashboards (4.16, except FR-17.3); AI classification (4.17); semantic matching, ranking, explanations and confidence (4.18–4.19); basic generative reports (4.22); AI tracking, overrides and fallback (4.23); sustainability indicators and dashboard (6.1); synthetic data and documented local setup (DR-01, C-07)
Desirable extensions (Should)	Sustainable-action recommendation (4.20); campus assistant (4.21); configurable approval values (FR-03.3); report export (FR-17.3); hosted deployment (DEP-02); richer sustainability dashboard
Future enhancements (Could / Won't this semester)	Email notifications (FR-16.6); demand forecasting; map view; saved searches and alerts; Arabic interface; live KU ERP, finance, procurement or identity integration; barcode or RFID hardware; payments; native mobile apps; custom model training

6.5 Traceability

Table 6.5 links each feature to its Phase 1 requirement, its use cases and the roles involved, so we can see that nothing from Phase 1 was dropped. Our Phase 7 test cases will use the same IDs.

Table 6.5 Feature traceability

Feature (section)	Phase 1 IDs	Use cases	Main roles
Accounts and access control (4.3)	FR-01, FR-02	UC-01	Business administrator, System administrator
Departments, categories and settings (4.4)	FR-03	UC-01	Business administrator
Asset registration and records (4.5)	FR-04, FR-05	UC-02	Asset custodian, Department representative
Resource publication (4.6)	FR-06	UC-03	Department representative, Asset custodian
Resource requests (4.7)	FR-07	UC-04	Requester, Department representative
Search and filtering (4.8)	FR-08	UC-05	Requester
Reservations (4.9)	FR-09	UC-07	Requester, Asset custodian
Approval workflow (4.10)	FR-10	UC-08	All approving roles
Transfer, handover and receipt (4.11)	FR-11	UC-09	Department representatives, Asset custodians, Finance and Procurement officers
Inspection, maintenance and repair (4.12)	FR-12, FR-13	UC-10	Asset custodian, Maintenance staff, Department representative, Finance officer
End-of-life actions (4.13)	FR-14	UC-11	Sustainability officer, Finance officer, Department representative
History and audit (4.14)	FR-15, FR-18	UC-14	Asset custodian, Business administrator
Notifications (4.15)	FR-16	All (included in UC-08)	All roles
Conventional reports (4.16)	FR-17	UC-13	Business administrator, Finance, Procurement and Sustainability officers

Feature (section)	Phase 1 IDs	Use cases	Main roles
AI classification (4.17)	AI-01	UC-02	Asset custodian
Semantic matching, ranking and explanations (4.18–4.19)	AI-02 to AI-05	UC-06	Requester
Sustainable-action recommendation (4.20)	AI-06	UC-11	Sustainability officer
Campus assistant (4.21)	AI-07, AI-08	UC-12	All roles
Generative reporting (4.22)	AI-09, AI-10	UC-13	Report data owners
AI tracking, overrides and fallback (4.23)	AI-11 to AI-13	UC-06, UC-12	All roles
Sustainability indicators (6.1)	SU-01 to SU-08	UC-13	Sustainability officer, Finance officer

Appendix A: Glossary

Term	Meaning
Asset	Anything the university owns that we track, or a batch of identical items with a quantity
Listing	An item a department has offered to other departments
Request	What a department needs, written as a form
Reservation	Holding some units of a listing for a limited time; not an approval
Transfer	Moving an item from one department to another, after approval
Custodian	The person responsible for an item's condition and location
Department representative	The person who approves listings, requests, receipts and spending for a department
Approval task	One decision waiting for someone with the right role (section 4.2.8)
Work order	A repair or maintenance job, from report to completion
End-of-life action	What happens to an item that leaves service: keep, donate, take apart, recycle or dispose
Compatibility score	How well an item fits a request, from 0 to 100; not a probability and not an approval
Confidence	How sure the AI is about its answer, shown separately from the score
Eligibility check	The normal rules that remove items that can't be used before the AI ranks the rest (BR-14)
Avoided purchase	Something the department didn't have to buy because it got an existing item (BR-18)
Waste diverted	Items kept out of the bin through reuse, repair, donation, taking apart or recycling
kg CO2e	Kilograms of carbon dioxide equivalent, the unit for carbon estimates
EARS	The "When / While / If … then / Where" way of writing requirements from the Chapter 3 slides
EHS	Environment, Health and Safety, the KU office we check with for special items
TBD	Something we still need the instructors to confirm (Appendix C)

Appendix B: Analysis Models

For this phase, our analysis models are the use-case diagram and descriptions in section 4.1 and the state tables in section 4.2. The other diagrams come later, as the project description sets out: context and data-flow diagrams in Phase 4, and the UML class, sequence, activity and state diagrams, the ER diagram and the AI workflow diagram in Phase 5. They will use the same IDs as this document.

Appendix C: To Be Determined List

ID	Open question	Our proposal	Used in
TBD-1	Item value at which a transfer also needs Finance approval	KU Finance to set	BR-06, FR-11.2
TBD-2	Repair cost at which Finance approval is also needed	KU Finance to set	BR-07, FR-13.2
TBD-3	How long a reservation is held	5 working days	BR-12, FR-09.3
TBD-4	How old an item's last check can be before it must be checked again	90 days	BR-16, FR-06.4
TBD-5	When an approval counts as overdue	3 working days	FR-10.5
TBD-6	Where the carbon factors for each category come from	A published factor set, to agree	SU-07, DEP-03
TBD-7	How long records and audit entries are kept	Same as KU's record policy	FR-18.4, DR-03
TBD-8	Which categories need special handling, and the EHS steps	To confirm with EHS	BR-09, FR-14.4
TBD-9	Who approves when a department has only one representative	Named by the Business administrator	BR-02
TBD-10	Which KU office would own the system	Facilities Management (from Phase 2)	2.1

Appendix D: Team Contributions and AI Use

Member	Sections	Branch
Sultan Almheiri (100065654)	1; 2.1–2.2; 2.5–2.7; 4.2–4.16; 5.5; 6; Appendices A–D; document build	phase3/sultan-requirements
Zayed Alfadli (100064657)	2.3; 3.1; 4.1	phase3/zayed
Ghaith Alhinaai (100066185)	2.4; 3.2–3.4; 4.17–4.23; 5.1–5.4	phase3/ghaith

AI use. We used AI tools to help prepare this document, build it from our files and organise the repository. Each of us reviewed our own sections and is responsible for their content.